Water Quality Study of Ladysmith Harbour, British Columbia

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WATER QUALITY STUDY OF LADYSMITH HARBOUR, BRITISH COLUMBIA

by

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ABSTRACT

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A bioassay using Pacific oyster Crassostrea gigas, Thunberg, embryos was done on surface waters collected from a depth of 0.5 m at five locations from the mouth to the head of Ladysmith Harbour, British Columbia from January 8 to September 17, 1979. Water quality as determined by the bioassay was poorer in Ladysmith Harbour water than in control water at the Pacific Biological Station in Nanaimo. There was a gradient in water quality in the harbour from the mouth, where it was best, to the head, where it was poorest but local exceptions to this trend were evident. Water samples were collected in 1979 and 1980 and analysed by gas chromatograph-mass spectrometry to identify the possible causes of the poor water quality. No definitive cause was identified but wood extractives, chlorophenols, fuel additives and dimethyl sulphide, all of which can be toxic to oyster larvae, were found. Six samples of water were collected from June 26 to September 13, 1979 and analysed for copper, mercury and arsenic content but concentrations of these metals were too low to explain the poor water quality.

Key words: Ladysmith Harbour water quality, oyster embryo bioassay, mass spectrometry, chlorophenols, wood extractives.

RÉSUMÉ

Bourne, N., H. Rogers, H. Mahood, and D. Neil. 1981. Water quality study of Ladysmith Harbour, British Columbia. Can. Tech. Rep. Fish. Aquat. Sci. 1026: 57 p.

Du 8 janvier au 17 septembre 1979, à cinq endroits compris entre l'entrée et le fond du port de Ladysmith (Colombie-Britannique), on a effectué un test biologique des eaux superficielles, recueillies à une profondeur de 0,5 m, en se servant d'embryons d'huîtres du Pacifique (Crassostrea gigas, Thunberg). D'après les résultats du test, l'eau du port de Ladysmith était de moins bonne qualité que celle servant de témoin à la station de biologie du Pacifique (Nanaimo). La qualité de l'eau variait graduellement depuis l'entrée, où elle était la meilleure, jusqu'au fond du port, où elle était la plus mauvaise, certains endriots échappaient cependant à cette règle. Des échantillons d'eau ont été recueillis en 1979 et 1980 et analysés par la méthode chromatographie en phase gazeuse-spectrométrie de masse afin de déterminer ce qui peut causer la mauvaise qualité de l'eau. Aucune cause précise n'a été identifiée, mais on a découvert la présence de substances toxiques pour les larves d'huîtres: produits d'extraction du bois, chlorophénols, additifs de carburant et sulfure de diméthyle. On a recueilli, du 26 juin au 13 septembre 1979, six échantillons d'eau et on a analysé leur teneur en curvre, en mercure et en arsenic, mais la concentration de ces métaux était trop faible pour expliquer la mauvaise qualité de l'eau.

Mots-clés: qualité de l'eau du port de Ladysmith, test biologique d'embryons d'hiîtres, spectrométrie de masse, chlorophénols, produits d'extraction du bois.

INTRODUCTION

Ladysmith Harbour is a narrow inlet on the lower east side of Vancouver Island approximately 8 km in length and varying in width from 0.7 to 2 km (Fig. 1). The harbour is divided into an inner shallower portion and an outer wider deeper part by a constriction approximately in the centre. There are extensive intertidal flats in the upper part of the inner harbour. Maximum water depth in the inner harbour is about 9 m and in the outer harbour about 37 m.

The harbour has been an important oyster producing area for many years. Originally native oysters Ostrea lurida Carpenter, were harvested there. Eastern oysters, Crassostrea virginica Gmelin, were introduced around the turn of the century but the transfer was not successful and the practice was discontinued. In 1912 or 1913 the Pacific oyster, C. gigas Thunberg, was introduced into the harbour and this started a flourishing industry. By 1925 limited Pacific oyster breeding had occurred and significant introductions of seed from Japan began in that year and continued for many years. Limited breeding was reported in 1926, 1930 and 1931 and the first major breeding occurred in 1932 (Quayle 1969). This was the beginning of a series of breedings of some magnitude in the harbour. Another major breeding occurred in 1936. The harbour became an important oyster growing area and in 1949 the British Columbia Provincial government established an oyster research laboratory there where studies on oysters and oyster culture were carried out until the facility was closed in 1957.

Oyster harvesting in the harbour ceased in June 1965 when it was closed to harvesting because of sewage contamination (Cooper and Kay 1975). An improvement was made in sewage treatment by the town of Ladysmith and, as a result of further water quality surveys in 1970 and 1975, part of the inner harbour was re-opened for direct culture and harvesting of oysters. All the outer portion of the harbour remained closed.

Despite the re-opening of much of the oyster growing area of the inner harbour, oyster production from Ladysmith Harbour declined and has virtually ceased. This was blamed on a deterioration in water quality that affected oyster growth and fatness as well as breeding. Breeding was particularly affected. The last commercial set of Pacific oysters in Ladysmith Harbour (mean of 10 spat per piece of Pacific oyster shell cultch) occurred in 1971 (Bourne 1978) although light scattered breeding occurred in some years since then. Hydrographic conditions since 1971 have been favorable in some years to produce a commercial set in the harbour (particularly 1975, 1977 and 1978). Spawning was observed and sufficient quantities of larvae to the early umbone stage were found in plankton to produce a commercial spatfall but none occurred (Bourne, unpublished MS; Heritage and Bourne 1979) (Table 1).

A further indication of the degradation of water quality was an apparent decline in the faunal species of the inner part of the harbour (D. B. Quayle, pers. comm.). Changes in the aquatic community structure are a well known indicator of altered water quality (Cairns and Dickson 1971; Cairns et al. 1972; Leathem et al. 1973). Most forms of stress reduce the complexity of aquatic ecosystems (Cairns and Dickson 1971; Boesch 1972).

Bottom fauna vary greatly in their sensitivity to various types of pollution. The introduction of a pollutant reduces the number of species by eliminating those most sensitive to the pollutant.

The apparent decline in water quality in Ladysmith Harbour caused serious hardships to the local oyster industry. Concern was expressed over the degree of deterioration of water quality, the cause or causes of this degradation and whether the cause(s) could be removed and the harbour water cleaned up to permit a return to former levels of oyster production.

In November 1973 the Provincial Land Management Branch, Department of Environment undertook a project to develop functional guidelines for management of the land and water areas and intrinsic natural resources of Ladysmith Harbour (Anon 1976). This comprehensive study of the harbour listed the various users, assessed conflicts between user groups and proposed guidelines for consideration for future management. A necessary step in formulation of the management plan was a determination of present water quality of the harbour and if degradation had occurred to make proposals for restoration to permit the oyster industry to attain former levels of production.

The Provincial Marine Resources Branch, Ministry of Environment and the Lands Branch, Ministry of Lands, Parks and Housing agreed to sponsor an oyster embryo bioassay project to assess water quality in the harbour. The project was carried out at the Pacific Biological Station in Nanaimo and began in September 1978 and terminated in December 1979. The bioassay would indicate the degree of water degradation but would not determine the source of the problem. Consequently arrangements were made with the West Vancouver Laboratory to undertake analyses of water samples from the harbour for organic compounds that could be the source of any water quality degradation.

In 1980 further studies were made to determine possible sources of water degradation. This project began in May and terminated in August and all analyses of water samples were again made at the West Vancouver Laboratory.

OYSTER EMBRYO BIOASSAY METHOD

The oyster or bivalve embryo bioassay technique has been used extensively to assess water quality in estuarine waters, (Anon 1980; Loosanoff and Davis 1963; Dimick and Breese 1965; and Woelke 1972). It is a sensitive and rapid indicator of water quality because developing bivalve embryos and larvae respond quickly to minor quantities of pollutants. During the initial 48-hour period of development, fertilized eggs normally develop into free-swimming, fully shelled, veliger larvae. Failure to develop into normal larvae or developing in an abnormal manner is used as an indicator of polluted or stressful conditions. The principal criterion has been a comparison of the percentage of normal larvae found in control and in test waters 48 h after fertilization. A further advantage of the test is

that embryos of many bivalves can be obtained on demand. The detailed method of the bioassay has been described in the above references but a brief outline follows.

During the spawning season adult oysters can be readily spawned in the laboratory. Out of the spawning season, adult bivalves are conditioned by gradually raising the water temperature until spawning temperatures are attained and then holding them at this temperature for 4-6 weeks until the gonads become mature. When embryos are required, adults are spawned by rapidly raising and lowering the water temperature at half-hour intervals in spawning dishes. The addition of a small amount of sperm extract is frequently used to initiate spawning if this is not achieved by temperature manipulations alone.

The spawned eggs are washed and filtered to remove pieces of tissue, resuspended in clean salt water and fertilized by adding small amounts of sperm. The number of zygotes is counted and a known number added to the test and control waters. Cultures are held for 48 hours at incubating temperatures which depend on the species of bivalve used; for Pacific oysters the temperature is usually 25°C. After 48 hours the larvae are filtered, resuspended in salt water and examined microscopically. A measure of water quality is obtained by comparing the number of normal veliger larvae with the total number of larvae or by counting the number of normal larvae that developed from the number of embryos in the control and test waters.

LADYSMITH HARBOUR BIOASSAY

SAMPLING SITES

To determine if a gradation in water quality exists in Ladysmith Harbour, five sampling sites were established which extended from the head (Station 1) to the mouth (Station 5), (Fig 1). All water samples were collected from a depth of 0.5 m to avoid surface contamination. Water samples were collected in glass containers that had been thoroughly cleaned and transported to the laboratory as soon as sampling was completed.

CONDITIONING AND SPAWNING ADULTS

Pacific Oysters, *C. gigas* were used for the bioassay since they are the species of commercial interest in the area, are easily conditioned and spawned and have been used extensively in other bioassay work.

Adults were obtained from six locations, three in the Ladysmith Harbour area, off Ivy Green Park, Northwest of Dunsmuir Island and off Coffin Point, and from three areas outside Ladysmith Harbour, Pipers Lagoon near Nanaimo, Nanoose Harbour and Henry Bay in Baynes Sound. Most of the bioassays were done with the Baynes Sound stock.

Adults were brought to the Pacific Biological Station laboratory and scrubbed to remove fouling organisms, placed in conditioning trays with running salt water and held at ambient water temperatures for two days. The temperature was raised 1°C every other day until 21°C was reached. Adults were held at 21°C ± 1°C for 3-4 weeks. Periodic samples were taken to determine the state of gonadal development. After the adults were held for 3-4 weeks at 21°C the water temperature was reduced to 19°C to prevent accidental spawning.

No difficulty was experienced in conditioning adults except those from Ivy Green Park. Those animals were in poor condition and produced very few eggs or sperm.

SPAWNING

When embryos were required for bioassays, adults, usually two per dish, were placed in shallow pyrex dishes in a water bath. Temperature of the water in the dishes was quickly raised to approximately 30°C and held there for half an hour after which it was quickly reduced to approximately 8°C and held there for half an hour. Subsequently the temperature was alternated between 30°C and 8°C at half hour intervals. Three or four such temperature manipulations were usually sufficient to trigger spawning. In some instances sperm extract was added and the combination of temperature fluctuations and addition of sperm extract triggered spawning. Difficulty in spawning adults was only experienced in one bioassay.

The eggs were washed through a 235 micron screen to remove excess gonadal tissue and resuspended in salt water at the incubating temperature. Three to four pipettes of sperm were added and the water thoroughly stirred. Fertilized eggs were examined microscopically and when the first polar body appeared the egg suspension was stirred and the number of eggs counted. Triplicate counts were made in a Sedgewick-Rafter cell.

BIOASSAY PROCEDURE

Test water from Ladysmith Harbour was filtered through a 37 micron screen to remove detritus. One liter aliquots from each sample were placed in three beakers, as each bioassay was done in triplicate. The beakers were placed in a water bath and the temperature of each sample brought to the incubating temperature.

Water for the control was from the Pacific Biological Station salt water system which has been tested by the oyster embryo bioassay method on several occasions and found to be of high quality (Bourne, unpublished MS). The inlet for the salt water system is approximately 20 m below the 0 tide level and 200 m from shore. Water for the control was also filtered through the 37 micron screen and one liter aliquots placed in three beakers and the temperature raised to incubating temperatures. All equipment used in the bioassays was tested by the bioassay technique to ensure it was non-toxic.

Fertilized eggs were added to each beaker to give a concentration of 20 per ml. The beakers were covered and left for 48 hours at incubating temperatures.

Originally the incubating temperature was 25°C. However, early in the work it became necessary to hold the beakers in a bath of heated salt water from the Pacific Biological Station's supply. This maintained cultures at a temperature of 23°C ± 0.5°C which proved satisfactory.

After 48 hours the larvae were filtered on a 37 micron screen, resuspended in sea water and examined microscopically for general condition and gross abnormalities. The larvae were resuspended in salt water, thoroughly mixed and a sample taken for determination of the percentage of normal larvae in the culture. When resuspended, the concentration was adjusted so that at least 50 larvae were examined in a 1 ml sample whenever possible. The percentage of normal larvae was calculated from the mean of three samples.

RESULTS AND DISCUSSION

Twenty-one bioassays were carried out on Ladysmith Harbour waters during the period January 8 to September 17, 1979 (Table 2). Results of these bioassays are summarized in Fig. 2-12.

The commonest type of abnormal larvae observed in the bioassays were those with misshapen valves; the ventral margin was crinkled or uneven, evoluted outward, and in some instances notched. Larvae with these conditions could not close the valves properly. Frequently the tissues of abnormal larvae were withdrawn and did not extend to the ventral margin of the shells. Tissues of some larvae were extruded and could not be withdrawn into the valves. The velum of some larvae was misshapen and did not extend from the margin of the valves. With these and other larvae, swimming was atypical and the larvae were obviously abnormal. In some instances very abnormal larvae developed (so called ear muffed or black spot). Another criterion was the colour of the larvae; normal larvae are brown in colour with a pinkish tinge, abnormal larvae lacked the pinkish tinge and were pale and washed out.

In most of the bioassays the number of larvae that developed in the 48-hour period and the percentage of normal larvae were higher in the control than in the test samples. There were two exceptions, those of January 22 and February 19. In both these bioassays the thermoregulator controlling the temperature in the bath malfunctioned and water temperatures varied from 15 to 31°C during the 48-hour period. After the February 19 bioassay all beakers were kept in a bath that was surrounded by water from the heated sea water system of the Pacific Biological Station. A further complication in the January 22 bioassay was that insufficient eggs were obtained from spawning adults and eggs were stripped from adult females. Results of these two bioassays may not be comparable with those of other bioassays.

In the January 8 bioassay the control water had 90% normal larvae while in the test water from Ladysmith Harbour the percentage of normal larvae ranged from 70-77% (Fig. 2). One sample from Station 1 had few larvae and was omitted from the calculations. There was no indication of any gradation in the number of normal larvae in the water samples. In the second assay on January 22, the control water had 64% normal larvae, but at Stations 3 and 2 the percentage of normal larvae was 81 and 83% respectively (Fig. 2). Only at Station 1 was the percentage of normal larvae lower than in the control. As discussed, the number of larvae in this assay was much reduced and results of the January 22 bioassay may not be comparable with those of other bioassays.

Larval numbers were lower in the single bioassay in February which may have been due to temperature fluctuations during the 48-hr incubation period (Fig. 3). The percentage of normal larvae in the control and test water from Stations 5-3 inclusive was similar but lower at Station 2 and much lower at the head of the harbour.

A gradation in the percentage of normal larvae was evident from the mouth to the head of the harbour in the two bioassays in March. On March 5 the percent normal larvae ranged from 83% at Station 5 to 72% at Station 3 and was considerably lower at Stations 1 and 2 (Fig. 3). The overall quality of the larvae at Station 1 in this bioassay was poor. On March 19, the percentage of normal larvae was again high in water from Station 5, dropped at Station 4, ranged from 49-42% at Stations 3 and 2 and was much lower at Station 1 (Fig. 4).

In April a similar gradation in the percentage of normal larvae was evident in all three bioassays (Fig. 4, 5). In the April 11 bioassay, the percentage of normal larvae was considerably higher at Station 2 than at Stations 3 and 4. This indicates that nonhomogenous conditions in water quality may exist in the harbour which could affect the percentage of normal larvae in the assay at any one location. In the other two April bioassays, the percentage of normal larvae at Station 2 was lower than at Station 3. The percentage of normal larvae at Station 1 was much lower than at the other Stations in all three bioassays indicating poor water quality at the head of the harbour.

A gradation from the mouth to the head of the harbour was again evident in the May 14 bioassay (Fig. 6). Numbers of larvae and the percentage of normal larvae were again low at Stations 1 and 2. In the May 29 bioassay the percentage of normal larvae was lowest at Station 5 (27%) and only 30% at Station 2, 43 and 47% at Stations 1 and 3 and 63% at Station 4. Again this indicates that water of poor quality may be either general or localized in the harbour. Water samples for this bioassay were collected near the end of an ebb tide (Table 1) which may have influenced water quality at the mouth.

The situation in the June 4 bioassay (Fig. 7) was similar to that of May 29. The lowest percentage of normal larvae was observed in water from Station 5, a slightly higher percentage at Station 2, about the same percentage at Stations 1 and 3 and the hightest percentage at Station 4. These water samples were collected on a flood tide and again may indicate local noxious conditions in the harbour. On June 20 a gradation from the mouth to the head of the harbour was again evident with the highest

percentage at Station 5 and the lowest at Station 1. The percentage of normal larvae was low at Station 3.

No general trend in the percentage of normal larvae was observed in the July 6 bioasssy (Fig. 8). The highest percentage of normal larvae was found at Station 3 followed by Station 2. There were very few larvae in water samples from Station 4 and those that were found were mostly abnormal. In the other two bioassays in July (Fig. 8, 9), a gradation was evident with low percentages at the head of the harbour and only slightly higher at Station 2.

Results of the four bioassays in August are somewhat mixed (Fig. 9, 10, 11). There was a gradation in the percentage of normal larvae from the mouth to the head of the harbour except for the bioassay of August 14. The percentage of normal larvae was very low at the head of the harbour and generally not much higher at Station 2. Except for the bioassay on August 20, the numbers of larvae and the percentage of normal larvae were lower at all Stations than observed in other bioassays. No explanation can be given for this general lowering of water quality in Ladysmith Harbour at that time. It agrees with observations made during our oyster breeding work in Ladysmith Harbour. During August the number of Pacific oyster larvae found in the plankton frequently drops markedly and may result in virtually no spatfall.

A gradation in the percentage of normal larvae was evident in the two bioassays in September (Fig. 11, 12), the percent normal larvae was low at Station 1 and only slightly higher at Station 2.

Sudden decreases in salinity are known to affect developing Pacific oyster embryos and the number of larvae (Woelke, 1972). Salinity measurements were taken when water samples were collected until the end of April with a YSI (Yellow Springs Instrument) Model 33 in situ salinometer. After that date they were inadvertently discontinued (Table 3). The salinity of the Pacific Biological Station salt water is 28‰ $\pm$ 1‰, on an annual basis. Salinities measured from January 8 to April 30 did not go below 23‰, (Table 3). McAllister (1956) stated that surface salinities ranging from 13‰ to 29‰ had been observed over a period of years in Ladysmith Harbour and there were three salinity minima each year. Two were associated with the autumn and winter peaks of fresh-water discharge. The third occurs in midsummer, when drainage has a negligible effect. However, the surface salinities he measured in the summer never fell below 22‰. Salinity measurements have been taken regularly in our Pacific oyster breeding work in June, July, August and September, (Heritage and Bourne 1979; Bourne, unpublished MS) and the surface salinity has never fallen below 24‰. It is believed that salinities throughout the course of the bioassays were above 23‰ and had little if any adverse affect on developing embryos.

Another putative cause for the difference in percentage of normal larvae between the control and test waters may be differences in numbers of bacteria in waters from the two sources. Loosanoff and Davis (1963) have shown that certain bacteria may cause mortalities in bivalve larvae. In previous work (Bourne, unpublished MS) no problem was experienced with bacterial contamination in bivalve larval cultures maintained in salt water

from the Pacific Biological Station and it is believed the bacterial level in this salt water is low. Bacterial surveys have been undertaken on Ladysmith Harbour waters (Cooper and Kay 1975) but it is felt that bacterial levels found would not cause the differences in percentage of normal larvae observed between the control and test waters.

It is believed the differences in percentages of normal larvae observed between the control waters at the Pacific Biological Station and waters in Ladysmith Harbour reflect differences in water quality between the two areas and indicate a degradation in water quality in Ladysmith Harbour waters. Support for this view comes from other sources. The decline in faunal species in the harbour which is a result of changes in water quality (Cairns and Dickson 1971; Cairns et al, 1972; Leathem et al, 1973), the lack of commercial Pacific oyster spatfall in recent years and the reported fact that growth and fatness in oysters has declined greatly in the past ten years. The bioassays support the supposition that there has been a decline in water quality in Ladysmith Harbour.

The percentage of normal larvae and hence water quality was generally lower in summer months than in the winter and early spring. The percentage of normal larvae declined in May and was generally lower until September when it improved, particularly in the outer portion of the harbour.

There was a general gradient in water quality from the mouth of the harbour where it was best to the head where it was poorest. Local exceptions to this general trend were evident in some bioassays which probably indicates that some sources of pollution may be localized initially and then disperse with currents. Log barges are dumped at the mouth of the harbour (Station 5) which could produce local poor water quality conditions that would gradually dissipate with time. This could explain some of the poor water quality noted occasionally at Station 5 and even Station 4. A detailed and almost continuous bioassay would be necessary to determine if such situations do exist.

The cause of the gradient in water quality in the harbour is unknown. Considerable log debris and other organic accumulations are present in the inner part of the harbour (Anon 1976). Leachates from this wood could produce toxic conditions. Another possible cause of water degradation in the inner harbour could be the high BOD of these organic deposits, particularly during periods of high water temperatures. Noxious gases are easily disseminated into the surrounding waters when the sediment is disturbed (Sibert and Harpham 1979) and this could produce toxic conditions for bivalve larvae. Oceanographic studies of Ladysmith Harbour show the rate of flushing in the inner harbour is low. Approximately one third of the water is exchanged on any tidal cycle (McAllister, 1956). Toxic substances would tend to remain there for longer periods of time before dispersing than in the outer part of the harbour.

Brood stock for the bioassays was collected from six locations, (Table 2). An original intention of this work was to determine whether there were differences in development of embryos from adults collected at different locations, but because of practical considerations this proved to be impossible. No difficulties were experienced in conditioning or spawning

adults from the six areas except those from Ivy Green Park which were in such poor condition that few eggs were obtained from them. Observations made during the bioassays indicated there was little if any difference in the survival or percentage of normal larvae of embryos from the six locations in either the control or test waters. The tentative conclusion from these observations is there are no genetic differences between stocks, shown by the development of embryos in water at the Pacific Biological Station and Ladysmith Harbour.

In the present study water samples were collected only from the 0.5 m depth. With a more extensive assessment, water quality should be tested at a series of depths to determine if toxic conditions vary vertically. This is particularly important for water close to the bottom because of debris decomposition. Although Sibert and Harpham (1979) found little bark deposition from intertidal log storage in the Nanaimo estuary, considerable bark accumulation from log booming operations has been reported in Ladysmith Harbour (Anon 1976) and in Howe Sound (McDaniel 1973). Leachates from this bark could produce conditions toxic to bivalve larvae. Natural decay of the wood material could reduce oxygen levels and produce toxic gases. These toxic substances might remain close to the bottom and cause deleterious conditions. However, they could be brought to the surface by upwelling which might produce local toxic conditions according to current configurations.

Another possible source of pollution in the harbour is the sewer outfall from the town of Ladysmith. Prior to 1965 raw sewage was dumped into the inner harbour but in 1965 a new sewage treatment plant was constructed at Holland Bank with an outfall extending 876 m into the outer harbour to a depth of 19 m (Cooper and Kay 1975). This is close to Station 4 of the present study. A single dye study by Cooper and Kay showed the circulation pattern at depth took some of the dye northward towards Dunsmuir Island but most was southward towards Vancouver Island and along the shore towards Davis Lagoon. From this single study one would expect little of the discharge to enter the harbour or extend to the inner harbour, particularly during the summer when there can be thermal stratification. However, there have been reports of some sewage reaching the narrows (Station 3) of the harbour. Further, if upwelling occurred, it is possible that any toxic substances in the sewage could affect oyster larvae in the surface water, particularly in the outer part of the harbour. This might explain some of the low percentages of normal larvae found in water samples from Stations 4 and 5 in some of the bioassays.

Another possible cause of deterioration in water quality is suspended matter in the water column. Both Levings and McDaniel (1973) and McDaniel (1973) reported clouds of suspended material in the water column resulting from mine tailings and this cloud travelled at least 1.5 km from the source. During oyster breeding studies, microscopic fibrous material has been observed in surface plankton samples (Anon 1976; Bourne, unpublished MS). The source of this material is unknown but it has been observed both in the inner and outer parts of the harbour. Whether these fibers are toxic to bivalve larvae is unknown but whenever they have appeared there has been a marked decline in the number of Pacific oyster larvae in the harbour. No microscopic fibers were observed in water samples during the bioassay but they may have been present.

ORGANIC CHEMICAL ANALYSIS

INTRODUCTION

Results of the oyster bioassay indicated there had been a degradation in water quality in Ladysmith Harbour but it did not reveal the identity of any toxicants. Analyses of harbour waters, sediment and oysters were undertaken at the West Vancouver Laboratory to determine if organic pollutants, that might be toxic to oyster larvae, were present in Ladysmith Harbour waters.

COLLECTION AND PROCESSING OF WATER SAMPLES

Short aluminum columns (61 x 3.8 cm) were constructed and filled with Amberlite XAD-2 synthetic resin, manufactured by Rohm and Haas Inc. Field experiments were conducted to try and pump 760-liter samples of harbour water through the resin beds at various locations (especially at Station 2). Unfortunately practical problems arose for column pressures were too great and the resin particles could not be contained within the columns. We therefore resorted to collecting 25-liter water samples in glass containers with aluminum foil-lined lids, transporting them to the Pacific Biological Station, and percolating the water through large columns (91 x 4.4 cm) of precleaned XAD-2 resin by gravity flow. After passage of the water samples, the resin beds were stripped with methanol (2 l) and the methanol extracts analysed in the West Vancouver Laboratory. This was done by gas chromatography with a flame ionization detector (GC-FID) by methods described below. Results were inconclusive and variable because the water sample size was too small relative to the high dilution of the pollutants and because of problems in keeping glassware absolutely clean.

In November and December 1979, 25-liter samples of water were collected daily from Station 2 and percolated through one of the long columns of Amberlite XAD-2 synthetic resin. After a total volume of 850 liters of seawater had been passed through the column (this took about three weeks) the resin bed was stripped with 3 liters of methanol and the methanol extract analysed.

No samples were taken from January to May 1980 because of a lack of funding. Commencing June 4, other water samples were processed as described above and detailed in Table 4. Methanol extracts from the synthetic resin beds were analysed. Procedural blanks on two resin beds were prepared by processing Departure Bay water samples and checking for the absence of organohalogen compounds. In addition, column blanks were obtained before each water sample was processed by passing a 2-liter portion of methanol through the resin bed to check for any possible contamination.

The first large water sample and blank, processed during June 1980, was treated differently from the other samples in that the water was

acidified to pH 2.0 by addition of sulphuric acid before passage through the resin bed.

On June 11, 1980 a foam sample was collected beside a logboom near Ivy Green marina and analysed. This was the only occasion when foam was sampled.

COLLECTION OF SEDIMENT AND OYSTER SAMPLES

Six surface sediment samples were collected from locations at the head of the harbour on June 11, 1980 as summarized in Table 5. These varied in texture from gravel containing considerable amounts of bark and wood debris (Sample 1) to fine silt. Sample 2, taken in the area of Ivy Green Park, contained decaying leaf litter and had a strong odour of hydrogen sulphide, presumed to indicate anaerobic conditions.

Adult oysters were collected on the foreshore near Ivy Green Park on November 29 and December 10, 1979 and June 4, 1980. Oysters were also collected at Henry Bay in Baynes Sound on November 30, 1979.

CHEMICAL ANALYSIS OF WATER SAMPLES

Methanol extracts were evaporated to dryness using a Buchi rotary evaporator and the extracts methylated with diazomethane in preparation for analysis. The methylated extracts were transferred to vials in solution in methylene chloride and the volumes adjusted to approximately 1 mL.

Analysis of extracts was performed on a Hewlett Packard Model 5711A gas chromatograph equipped with a flame ionization detector (GC-FID). A 180 x 0.635 cm o.d. glass-walled column packed with 3% OV-101 on 80-100 mesh Supelcoport was used. The oven was maintained at 125°C for 2 min, then raised to 250°C at 6°C per min and maintained at 250°C for 1 hour.

Selected samples, dissolved in methanol, were analysed with a Hewlett Packard Model 5713 A gas chromatograph equipped with a Ni^{63} electron capture detector (GC-EC) for halogenated organics. The column was similar to that described above but the temperature was held constant at 150°C.

Analysis of extracts by gas chromatography-mass spectometry (GC-MS) was conducted on a Hewlett Packard Model 5992 A instrument equipped with a glass-walled packed column similar to that described above. The instrument was operated using a membrane separator and peaks were examined under electron impact at an ionizing voltage of 70 ev. Output of the instrument consisted of a total ionization profile (also known as a reconstructed gas chromatogram (RGC)) and the mass spectra of individual peaks were recorded after the elapse of various time intervals from the time of injection of the sample. Standard compounds available in the laboratory, were also injected into GC-FID, GC-EC and GC-MS instruments and their

retention times and mass spectra compared with the unknown compounds detected in the water extracts.

The foam sample (1500 ml) was acidified to pH 3 with sulphuric acid and extracted three times with methylene chloride. The combined ether extracts were dried over anhydrous magnesium sulphate, filtered and evaporated to dryness in a rotary evaporator. The residue was methylated in the usual way and dissolved in methanol. Analyses for pentachlorophenol and tetrachlorophenol were conducted by GC-MS in the single ion monitoring mode (SIM) using in the first case ions at m/e 282, 280, 278, 267, 265 and 263 and in the latter case the ions at m/e 248, 246, 244, 233, 231 and 229.

ANALYSIS OF SEDIMENT SAMPLES

Sediment samples were analysed according to a standard procedure (Renberg 1974). Duplicate (5 gm) portions of each sample were extracted with 20 mL and 10 mL aliquots of 0.2 M sodium hydroxide solution followed by centrifugation. Sephadex QAE-A25 ion exchange resin, (4 mL swollen volume) was placed in a centrifuge tube and 3 mL of the sodium hydroxide extract added. The tube and contents were shaken and centrifuged, the supernatent liquid being discarded. Distilled water was added to the resin, tube and contents shaken and centrifuged. The water layer was discarded. Benzene (3 mL) and acidic buffer (3 mL; 50:50 mixture of 0.2 M hydrochloric acid and 0.2 M potassium chloride) was added, tube and contents shaken and centrifuged. The benzene layer was drawn off with a pipette and evaporated to dryness prior to methylation. The extracts were analysed by GC-FID and GC-EC. One extract was also analysed by GC-MS.

ANALYSIS OF OYSTER TISSUE SAMPLES

In late November 1979 adult oysters were received from Station 2 in Ladysmith Harbour and Baynes Sound. One animal from each sample was homogenized in a Sorval mixer in 50 mL distilled water. A portion (15 mL) of this homogenate was added to 15 mL distilled water in the glass tube of a Hewlett-Packard 18961 A purge and trap sampler and the samples purged at room temperature using a stream of helium. The volatiles expelled in the purge gas were collected in a trap packed with Tenax-GC material. Subsequently the trap was rapidly heated and the volatile compounds desorbed into the GC-MS instrument where they were separated on a 180 x 0.635 cm o.d. glass-walled column packed with 0.2% Carbowax 1500 on 80-100 mesh Carbopack.

On December 10, 1979 another consignment of 13 freshly gathered oysters was received from Station 2 in Ladysmith Harbour. These were bulked using two large or three small animals to give five samples. Each sample was homogenized with 25 mL distilled water and 25 mL of the homogenate added to 10 mL of distilled water in the glass tube of the purge and trap apparatus. The purged volatile compounds were analysed as before.

On June 6, 1980 freshly shucked oysters from Station 2 were examined for the presence of chlorinated phenols. Half the sample was homogenized and mixed with anhydrous sodium sulphate to give a dry solid. This was soaked in cold methylene chloride and filtered. The residue was washed twice with methylene chloride and the bulked extracts evaporated. The residue was methylated and analysed by GC-EC and GC-MS in the SIM mode.

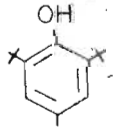
The other half of the sample was homogenized and heated in the presence of 20 mL of 1 N hydrochloric acid at 60°C for two hours. The residue was neutralized with sodium hydroxide and a methylene chloride extract prepared and analysed as described above.

RESULTS AND DISCUSSION

Twenty-five-liter water samples, processed during the summer and fall of 1979 gave inconclusive results in that extracts were heavily contaminated by trace chemicals being removed from the Amberlite XAD-2 resin or from the laboratory glassware. The GC-MS instrument can detect organic pollutants generally down to a concentration of 10^{-9} gm (1 nano gram) when operating in the peakfinder mode. When operated in the single ion monitoring mode (SIM) it can normally detect 10^{-10} gm (100 picograms). The problem of glassware contamination by traces of chemicals used in the laboratory is therefore a serious one and requires use of a high temperature oven in which glassware can be heated at 400 to 450°C to destroy such compounds and this was not available.

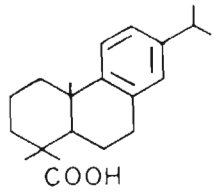
To counteract this problem we switched from processing 25-liter samples to 850-liter water samples. The flasks, traps and glass introduction pipes to the rotary evaporators were specially soaked in RBS-35 surfactant (Pierce Chemical Co) overnight, rinsed with distilled water and dried. These precautions effectively removed the contamination problem but meant that only a few water samples could be processed during the study period.

The RGC of a 5-liter water sample collected on October 29, 1979, contained a peak which gave rise to the mass spectrum shown in Fig. 13B. This is a rather simple spectrum of an aromatic compound which happens to be that of 2,6 di-tertiarybutyl-4-methylphenol (I). This compound is a common antioxidant used in the manufacture of automobile tires and is a gasoline additive. This compound most likely had its origin at a nearby marina where many tires, used as fenders, were exposed in the water. However, we have also found this substance in municipal sewage samples from the Vancouver area, so we cannot rule out some input from the Ladysmith Harbour sewage outfall.

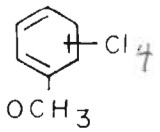
I 

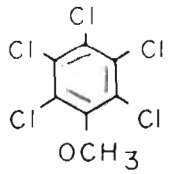
An 850-liter water sample processed in December 1979 yielded the most interesting and definitive results of the study. Organic pollutants present gave the mass spectra shown in Fig. 13A and the three in Fig. 14.

The compound giving rise to Fig. 13A is dehydroabietic acid (II), a resin acid present in softwood species and a compound known to be acutely toxic to young salmon in the 1 to 2 mg/l concentration range, (Davis and Hoos 1975). This acid always occurs together in wood with a number of other resin acids of related chemical structure. These are of lower water solubility than dehydroabietic acid and would consequently have a lower tendency to dissolve in saltwater if a piece of softwood were floating in the sea. Nevertheless, the presence of another acid (molecular weight 316 as the methyl ester) was inferred from the SIM (not shown) and as the compound had a lower retention time than dehydroabietic acid, the compound involved was most likely isopimaric acid. The origin of these two resin acids would probably be the logbooms in the harbour.

II 

The compounds in Fig. 14 were of special interest. Fig. 14B arises from tetrachloroanisole (III) and 14C from pentachloroanisole (IV). These

III 

IV 

are derived from tetrachlorophenol and pentachlorophenol respectively by reaction with diazomethane. We attempted to analyse the extract without methylation and were not able to detect pentachlorophenol because it is apparently too acidic to pass through the column and reach the detector. However, tetrachlorophenol was detected and two isomers could be discerned. After methylation of the samples, the two tetrachloroanisoles were apparently not resolved on this GC column. It should be noted that the four ion fragments with the asterisks in Fig. 14B arise from the fatty acid ester methyl laurate, which is a naturally occuring component extracted from algae and bacteria in the seawater and which emerges from the GC column simultaneously with tetrachloroanisole. The clusters of ions around the molecular ion M^+ in Fig. 14B and 14C arise from the chlorine atoms in these molecules because chlorine consists of a mixture of isotopes ^{35}Cl and ^{37}Cl in the ratio of about 3 to 1. Thus the molecular ion in Fig. 14B shows a four-chlorine pattern and that in 14C corresponds to a five-chlorine pattern. The satellite ion clusters below the molecular ion in each case show the loss of a methyl group and of an acetyl group, both typical of the fragmentation of phenolic methyl ethers. The instrument detection limits forthe pentachloroanisole were determined to be 5×10^{-9} g in the peakfinder mode and 1.2×10^{-10} g in the SIM mode.

In Fig. 14B we again note the clusters of ions around the molecularion and the loss of methyl and acetyl groups. This compound is also a phenolic methyl ether but the halogen in this case is bromine. The molecularion pattern is that associated with two bromine atoms - a mixture of nearly equal amounts of the isotopes ^{79}Br and ^{81}Br. The amount of dibromophenol in the extracts was minute and its presence is puzzling as we do not know of anyindustrial use for dibromophenol. It may be a natural product as a number ofbrominated terpenes have been identified as originating in marine algae (Faulkner 1977). Such compounds may be metabolized by animals which feed on the algae and there is a known marine

animal, Balanoglossus biminiensis, which has been found to excrete dibromophenol (Ashworth and Cormier 1967).

The origin of tetrachlorophenol and pentachlorophenol in Ladysmith Harbour is probably the sawmills. In British Columbia it is common practice to treat lumber for export markets with a mixture of tetrachlorophenol and pentachlorophenol to control sapstain fungi during shipment as these would reduce the saleability of the lumber on the export market. Pentachlorophenol is acutely toxic to rainbow trout and sockeye salmon in the range of 37-130 µg/l as the sodium salt (Davis and Hoos 1975) and is also known to be toxic to oyster larvae. Eastern oyster larvae, C. virginica, embryos exposed for 48 hr to the sodium salt of pentachlorophenol exhibited abnormal development. The measured 48 hr EC_{50} value was 40 µg/l (Borthwick and Schimmel 1978). Also similar observations have been made with Pacific oyster C. cigas larvae. The measured 48 hr EC_{50} level was 48 µg/l (Woelke, 1972). In a recent survey of eleven sites on the Lower Mainland and southern Vancouver Island, concentrations of pentachlorophenol in the range of 1 to 2100 µg/kg wet weight were measured in liver tissue of marine and freshwater sculpins (Anon 1979). Likewise tetrachlorophenol loadings of 29 to 1600 µg/kg wet weight were observed in sculpin liver tissue samples. Most of the sites studied adjoin lumber mills where pentachlorophenol is used as a fungicide, but Ladysmith Harbour was not included in the survey.

A foam sample was examined for the presence of organic pollutants. Analysis by GC-MS in the peakfinder mode gave inconclusive results. However, when the ions with m/e values of 263, 265, 267, 278, 280 and 282 were sought for in the foam extract, the result shown in Fig. 15 was obtained. These ions constitute the molecular ion cluster and the M^+-15 cluster from the mass spectrum of pentachloroanisole. Clearly this compound was detected by the instrument with a retention time of just over 8 minutes. Similarly in Fig. 16 we show the results of the search for tetrachloroanisole in the SIM mode. This compound was detected at a retention time of about 5.25 minutes.

All other samples of water, sediment and oysters analysed were negative for the presence of the two chlorinated phenols. The compounds might have been present in the 900-liter sample collected in the period June 4 through July 7, 1980. Unfortunately this water sample was made highly acidic before passage through the XAD-2 resin bed. Whereas it is true that some compounds have been shown to be more efficiently trapped by XAD-2 and XAD-4 resin under acidic conditions, pH 2.0 is too highly acidic and this resulted in the destruction of any organic compounds that may have been present and the isolation of a black oily residue from both the Ladysmith Harbour and Departure Bay water samples processed in this way. All other water samples were processed at their natural pH and without filtration.

Of the remaining water samples, the 450-liter runs collected July 23 through August 8, 1980 both contained small amounts of dehyroabietic acid and the sample taken at Station 2 also showed the presence of 2,6 di-tertiarybutyl-4 methylphenol (I). The two smaller samples collected in late August showed only the normal series of fatty acids observed in all environmental samples.

There were serious contamination problems associated with processing of 25-liter water samples in the early part of the study. Many of these artifacts arose from chemicals present in the XAD-2 resin and remaining from the manufacturing operation. These compounds can be subtracted from the RGCs by comparison with the spectra recorded on column blanks. Also artifacts that came from dirty glassware in the laboratory can be discounted through a knowledge of the types of compounds that were being handled in the laboratory at that time. What remains unexplained is a mystery compound with a base peak at m/e 45, an apparent molecular ion at m/e 200 and other ions at m/e 15, 41, 55, 59, 85, 169, and 172. This unknown compound was present in water samples collected in the harbour on August 13 and the m/e 45 peak is suggestive of a glycol group which would be used to confer water solubility. Unfortunately we have not analysed for additives in marine fuel sold at the marinas but this should be done if any further studies are contemplated.

Analysis of sediment samples did not reveal the presence of chlorinated phenols. It would be tempting to conclude from this finding that the sediments in the harbour are not acting as a sink for chlorinated phenols. However, more extensive sediment sampling should be performed and samples analysed from the deeper parts of the harbour before any conclusions can be reached on this matter.

Purge and trap analysis of homogenized oyster tissue samples for volatile organic compounds showed a difference between the animals collected in Ladysmith Harbour and those collected at a clean control site on Baynes Sound. The Ladysmith oysters contained dimethyl sulphide (Fig. 13C), whereas the control samples did not. This finding correlated with a "bad" odour noted when the Ladysmith oysters were freshly shucked. Apparently some phytoplankters can produce dimethyl sulphide as a metabolite (Ackman et al., 1966; Ishida, 1968). Whether the oysters obtained the compound from their food supply or directly from the water is unknown. It is possible that this finding may be another indication of oysters living under unhealthy conditions but in the absence of a much wider survey this cannot be supported by our present findings. Analysis for nonvolatile compounds in oysters collected in June 1980 at Ladysmith showed that chlorinated phenols were not present and only a series of saturated and unsaturated fatty acids was detected.

INORGANIC CHEMICAL ANALYSIS

Concern was expressed that poor water quality might be caused by high concentrations of heavy metals in the harbour waters. High copper concentrations have been recorded from oysters near the slag heap at Ladysmith and a solution of arsenic has been used sometimes to spray log booms to control shipworm attack (Quayle 1969). A brief study was made to assess levels of mercury, copper and arsenic in Ladysmith Harbour waters.

Water samples were collected at the 0.5 M depth at Station 2 on six occasions from June 26 to September 13, 1979. The samples were frozen

in 500 mL plastic bottles and sent to Cantest Limited, 1650 Pandora Street, Vancouver, B.C. for analysis. The method used in the analysis for the three metals was:-

Copper was determined using MIBK extraction and atomic absorption spectrophotometry

Arsenic was determined using hydride generator atomic absorption spectrophotometer

Mercury was determined using a Pharmacia Mercury Monitor (cold vapour atomic absorption)

Results of the analysis of the six samples are shown in Table 5.

Concentrations of the three metals were very low in all six samples, well below the LC_{50} limits for the embryonic development of the American oyster, C. virginica (Calabrese et al., 1973).

From these limited analyses, the conclusion is that concentrations of mercury, copper and arsenic are too low to have caused the poor water quality observed in the oyster bioassay of Ladysmith Harbour waters.

GENERAL DISCUSSION

Results of the bioassays indicate a deterioration of surface water quality in Ladysmith Harbour and corroborate similar conclusions from other observations; decline in the number of faunal species, recent lack of Pacific oyster breeding, and the poor growth and condition of adult oysters. There was a general gradation in the degree of water degradation from the mouth to the head of the harbour. Water quality was poorest at the head of the harbour and probably wouldn't support Pacific oyster breeding there. Although the best water quality was found at the mouth of the harbour, there were periodic occasions when it showed a marked depression which could have important consequences on oyster breeding.

Neither the organic nor inorganic chemical analyses indicated the cause or causes of the deterioration in water quality or ascribed which user groups are responsible for the present state of the harbour waters. The presence of dehydrobietic acid indicates log storage could be one of the sources of pollution. Tetra and pentachlorophenols were only found on isolated occasions but they could be important intermittent pollutants in the harbour. Derivatives from marine fuels could be another source of pollution, particularly if they occur in significant quantities. Dimethyl sulphide is another pollutant and could be a byproduct of logging activities. Bark would accumulate in areas where logs are stored and when this organic material decomposes, anaerobic conditions would occur which could be noxious to oyster larvae.

Some immediate steps could be undertaken to alleviate the state of pollution in the harbour. Tetra and pentachlorophenols probably come from local sawmills and steps should be taken to ensure these chemicals do not enter the harbour from local sources. Piles of wood waste stored above the high tide, particularly at the head of the harbour, should be removed since leachates from them are known to be toxic to fish (Peters et al., 1976; Pease 1974) and are probably toxic to oyster larvae.

Extensive studies are needed to determine the cause(s) of pollution in the harbour and the degree to which the various user groups are responsible for this pollution. However, cause and effect relationships would be difficult to quantitate.

A detailed oceanographic study of the harbour is needed with particular emphasis on currents both at the surface and at depth. Current pattern studies from the sewage outfall should be expanded to determine what effect this water has on other parts of the harbour. Oxygen levels at various depths should be measured over a period of time, particularly during summer when thermal stratification may occur, to determine if oxygen depletion occurs at depth. If oxygen depletion occurs, the influence of this water on surface waters should be assessed. Stress caused by the presence of sublethal levels of organic toxicants might result in mortalities in oxygen-depleted sea water.

The degree of toxicity of chemicals found in the present study on oyster larvae should be measured. A particularly important investigation would be a study of substances leaching from logs stored in sea water and their effect on oyster larvae, adult oysters and other marine organisms. This has far reaching implications since logs are stored in a variety of sites along the British Columbia coast and it is important to know the effect of this storage on the marine ecosystem.

The organic chemical analyses should be expanded and carried out on a routine basis. Quantification of chemicals in the harbour waters that are toxic to oyster larvae should be undertaken with particular emphasis on wood leachates. Derivatives of marine fuel products might cause some degree of pollution and should be investigated. Regular monitoring of the sewage outfall would indicate if significant quantities of toxic substances are continuously or intermittently entering the harbour. The BOD of the sewage should be determined since this might be critical during the warmer weather months when stratification occurs. Chemical studies of the sediment, particularly in the deeper areas, would show if any toxic materials are being incorporated there. The amount of chlorophenols in liver tissue of marine animals such as sculpins should be measured to determine if levels in the harbour are higher or lower than in other localities in the Province.

Ladysmith Harbour was an important oyster producing area and it is highly desirable that it return to its former levels of production. The amount of bottom suitable for intertidal oyster culture in southern British Columbia is limited to about 1,000 hectares (Quayle 1969) of which about 10% is in the Ladysmith Harbour area. Loss of this ground would be a serious impediment to the British Columbia oyster industry in its efforts to attain former levels of production.

An important question is, if the causes of pollution were immediately removed, would Ladysmith Harbour recover to permit the former type of marine community and if so how long would it require. No definite answer can be given but it should not prevent efforts to identify the causes of pollution in Ladysmith Harbour and to remove them to enable the harbour waters to recover and permit normal levels of oyster production.

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Table 1. Pacific oyster breeding in Ladysmith Harbour, British Columbia, 1912 - 1979.

1912-25	a few hundred (whole harbour)	1953	noncommercial (nil)
1926	a few hundred (whole harbour)	1954	nil
1927	a few dozen (whole harbour)	1955	nil
1928	a few dozen (whole harbour)	1956	commercial (25 per shell, 10 per shell)
1929	nil		
1930	a few hundred (whole harbour)	1957	nil
1931	a few hundred (whole harbour)	1958	commercial (several sets up to 500 per shell)
1932	commercial (13 per shell)		
1933	light noncommercial	1959	nil
1934	light noncommercial	1960	nil
1935	nil	1961	commercial (46 per shell, 30 per shell)
1936	commercial, 2 sets, 100 per shell		
1937	light noncommercial	1962	nil
1938	light noncommercial	1963	nil
1939	light noncommercial	1964	nil
1940	nil	1965	light
1941	noncommercial	1966	light (15 per shell)
1942	commercial (50 per shell)	1967	commercial (50 per shell)
1943	commercial	1968	light
1944	commercial	1969	
1945	noncommercial (2 per shell)	1970	
1946	nil	1971	commercial (30 per shell)
1947	noncommercial	1972	nil
1948	noncommercial	1973	nil
1949	commercial (10 per shell)	1974	noncommercial
1950	noncommercial (5 per shell)	1975	noncommercial light
1951	noncommercial (1 per shell)	1976	nil
1952	noncommercial (nil)	1977	noncommercial, (1 per shell)
		1978	noncommercial (less than 1 per shell)
		1979	nil

Table 2. Date water samples collected, state of tide and source of Pacific oyster broodstock for bioassays of Ladysmith Harbour waters, January 8 to September 17, 1979.

Bioassay No.	Date water sample collected 1979	State of tide	Source of brood stock
1	January 8	F	1,2,3,4
2	January 22	E	1,2,3,4
3	February 19	E	1,2,3
4	March 5	E	1,2,3
5	March 19	E	1,2,3
6	April 11	Slack → E	1,2,3
7	April 23	F	1,2,3
8	April 30	E	4,5,6
9	May 14	E	4,5,6
10	May 29	E	4,5,6
11	June 4	F	6
12	June 20	F	6
13	July 6	F	6
14	July 18	Slack → E	6
15	July 31	Slack → E	6
16	August 7	F	6
17	August 14	Slack → E	6
18	August 20	F	6
19	August 27	E	6
20	September 10	E	6
21	September 17	F	6

E Ebb tide
F Flood tide

Brood stock source

1. Ivy Green Park, Ladysmith Harbour
2. Northwest of Dunsmuir Island, Ladysmith Harbour
3. Coffin Point, Ladysmith Harbour
4. Pipers Lagoon, Nanaimo
5. Nanoose Bay
6. Henry Bay, Baynes Sound

Table 3. Surface salinities in Ladysmith Harbour, January 8 to April 30, 1979.

Depth (m)	Sampling station				
	1	2	3	4	5
			January 8, 1979		
0	28.7	28.7	28.6	28.8	28.8
1	28.8	28.8	28.7	28.9	29.0
2	28.6	28.8	29.0	28.9	28.9
3	28.5	28.9	28.9	29.0	29.0
			January 22, 1979		
0	27.6	27.9	26.8	28.5	28.6
1	28.8	28.7	28.8	28.9	28.8
2	28.8	28.8	29.0	28.9	29.0
3	28.8	28.9	28.9	29.0	28.9
			February 19, 1979		
0	23.3	23.6	23.5	27.9	28.5
1	26.7	28.0	28.1	28.2	28.5
2	28.8	28.5	28.6	28.7	28.9
3	28.8	28.8	28.7	28.7	28.8
			March 5, 1979		
0	24.6	24.4	25.1	27.1	27.6
1	26.6	26.9	26.7	27.3	27.9
2	27.6	27.7	27.5	28.1	28.1
3	28.3	28.2	28.3	28.2	28.5
			March 19, 1979		
0	26.3	26.5	26.6	26.7	27.1
1	27.1	26.9	27.3	27.5	27.8
2	27.8	27.3	27.6	27.9	27.7
3	28.0	27.8	28.1	27.9	28.3
			April 11, 1979		
0	26.9	27.2	27.4	27.5	27.5
1	27.1	27.3	27.4	27.6	27.5
2	27.2	27.5	27.5	27.8	27.7
3	27.8	27.8	27.6	27.7	27.9
			April 23, 1979		
0	26.0	26.8	26.8	27.0	27.0
1	26.3	26.9	27.1	27.1	27.2
2	26.5	27.0	27.3	27.1	27.4
3	26.9	27.3	27.2	27.2	27.6
			April 30. 1979		
0	27.2	27.1	27.0	27.3	27.3
1	27.3	27.1	27.2	27.3	27.4
2	27.3	27.1	27.1	27.4	27.1
3	27.4	27.2	27.3	27.4	27.4

Table 4. Water samples processed for organic pollutant analysis.

Dates collected	Sample type	Volume (liters)	Method of Concentration
October 29, 1979	Harbour water grab	25	XAD-2 resin
Nov. 30 to Dec. 14, 1979	Harbour water	850	XAD-2 resin
June 11, 1980	Foam near log boom	1.5	Methylene chloride extract
June 4 to July 7 1980	Harbour water	900	XAD-2 resin acidified to pH2.0
June 9 to July 8 1980	Departure Bay control	900	XAD-2 resin acidified to pH2.0
July 18, 1980	Blank on column #1	--	XAD-2 resin extractibles
July 18, 1980	Blank on column #2	--	XAD-2 resin extractibles
July 23 to August 8 1980	Harbour water Pacific Logging	450	XAD-2 resin
July 23 to August 8 1980	Harbour water Station 2	450	XAD-2 resin
August 20 to 27 1980	Harbour water Pacific Logging	250	XAD-2 resin
August 20 to 27 1980	Harbour water Station 2	220	XAD-2 resin

Table 5. Field samples of sediments submitted for chemical analysis.

Sample Number	Collection Date	Location	Notes
1	11 June 1980	Pacific Logging Spillway	Mixture gravel, bark and wood debris
2	11 June 1980	Ivy Green Park, foreshore west	Black mud with decaying vegetation. Rich in hydrogen sulphide
3	11 June 1980	Ivy Green Park, foreshore centre	Clean silt
4	11 June 1980	Ivy Green Park foreshore east	Clean silt with shell fragments
5	11 June 1980	Across from Ivy Green	Clean silt
6	11 June 1980	North end of Harbour across from sawmill	Clay with decaying leaf litter

Table 6. Concentration in parts per million (milligrams per liter) of copper, arsenic and mercury in water samples from the 0.5 m depth at Station 2 in Ladysmith Harbour from June 26 to September 13, 1979.

Date	Copper	Arsenic	Mercury
Detection Limit	0.001	0.001	0.0002
June 26	0.007	0.002	L 0.0002
July 4	0.003	0.001	L 0.0002
July 9	L 0.001	0.001	L 0.0002
July 17	0.003	0.001	L 0.0002
August 23	0.002	0.001	L 0.0002
September 13	0.003	0.001	L.0.0002

L = less than

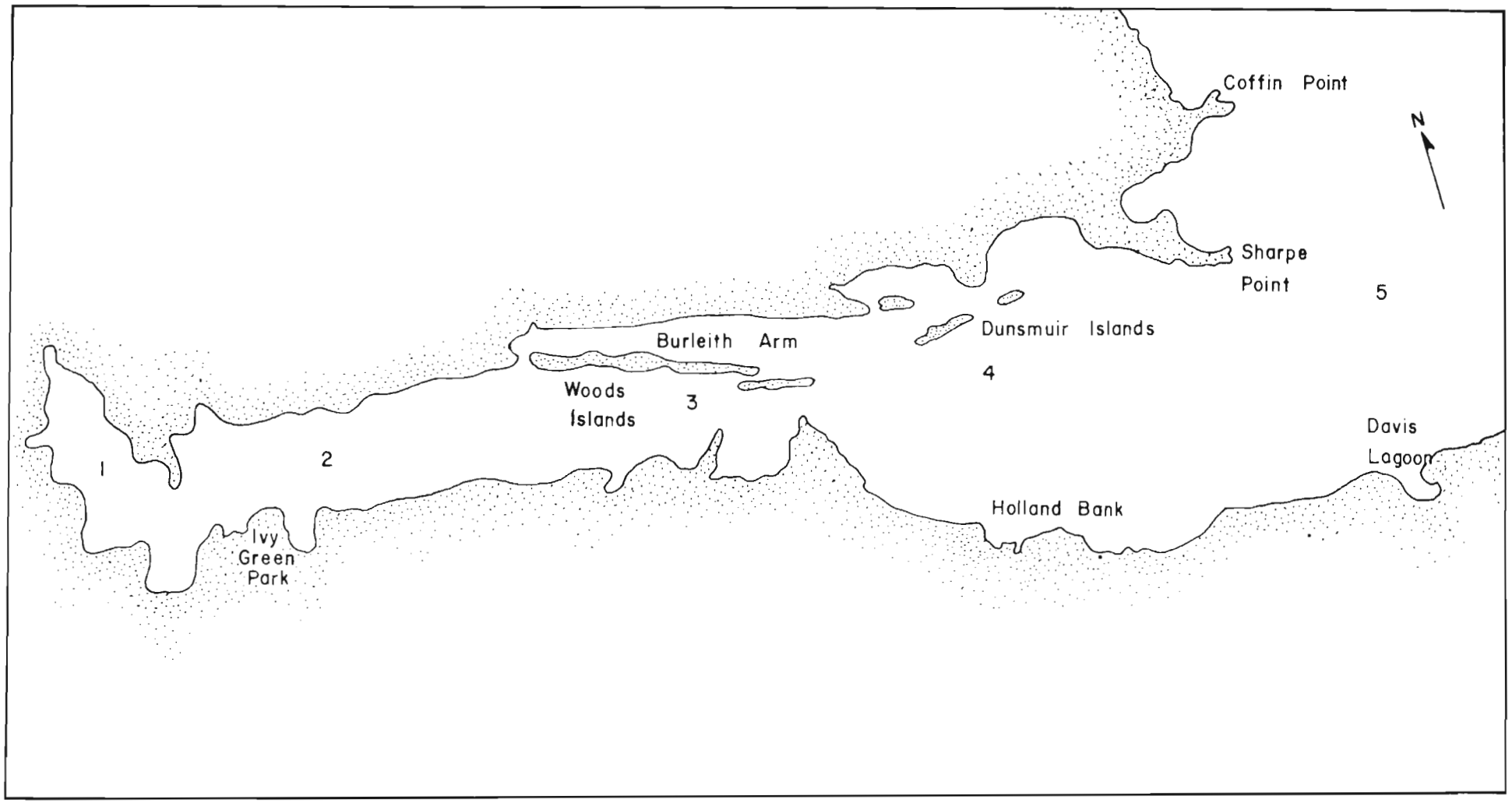


Fig. 1. Map of Ladysmith Harbour showing the location of the sampling stations.

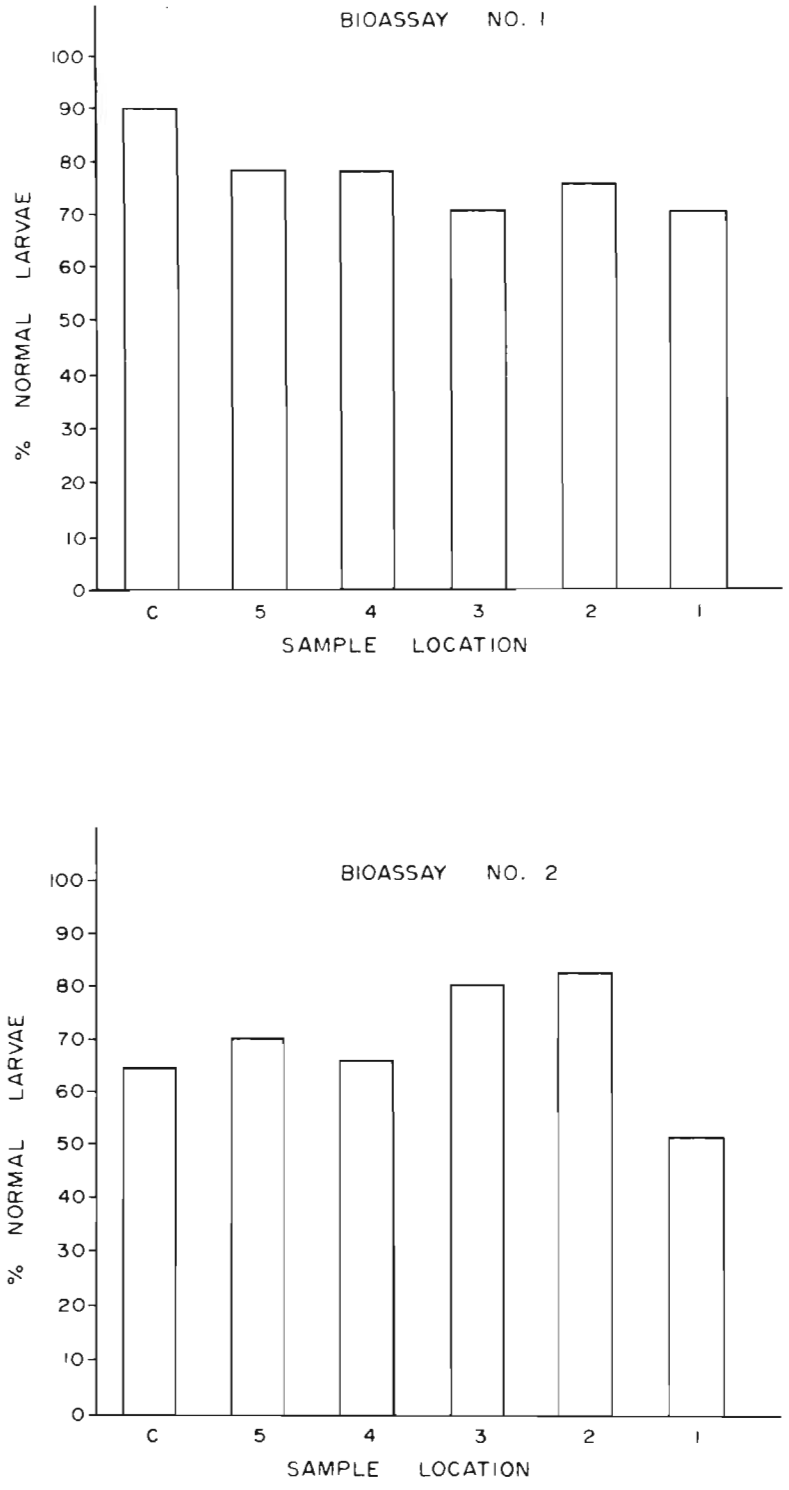


Fig. 2. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, January 8 above and January 22, 1979 below.

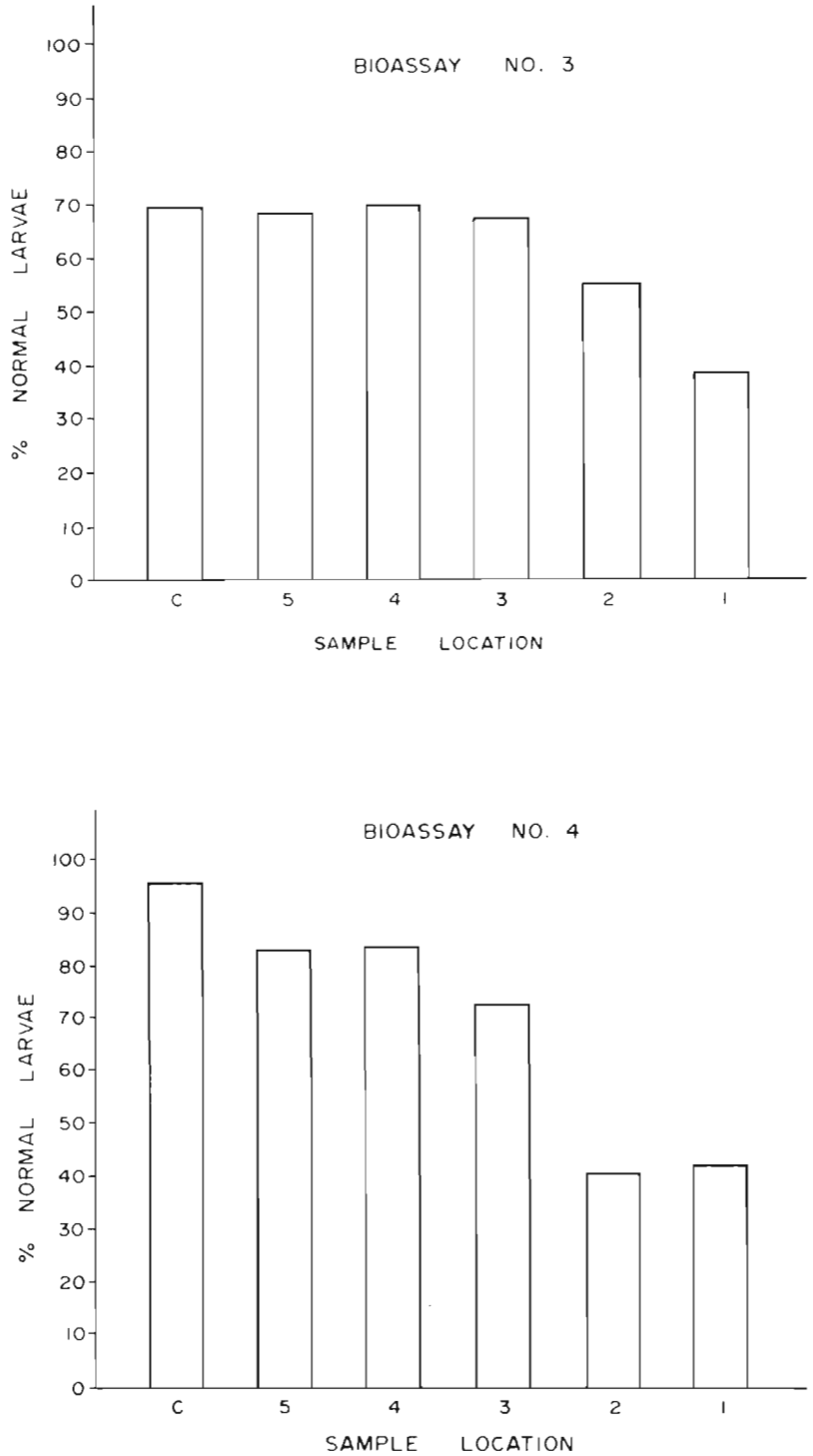


Fig. 3. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, February 19 above and March 5, 1979 below.

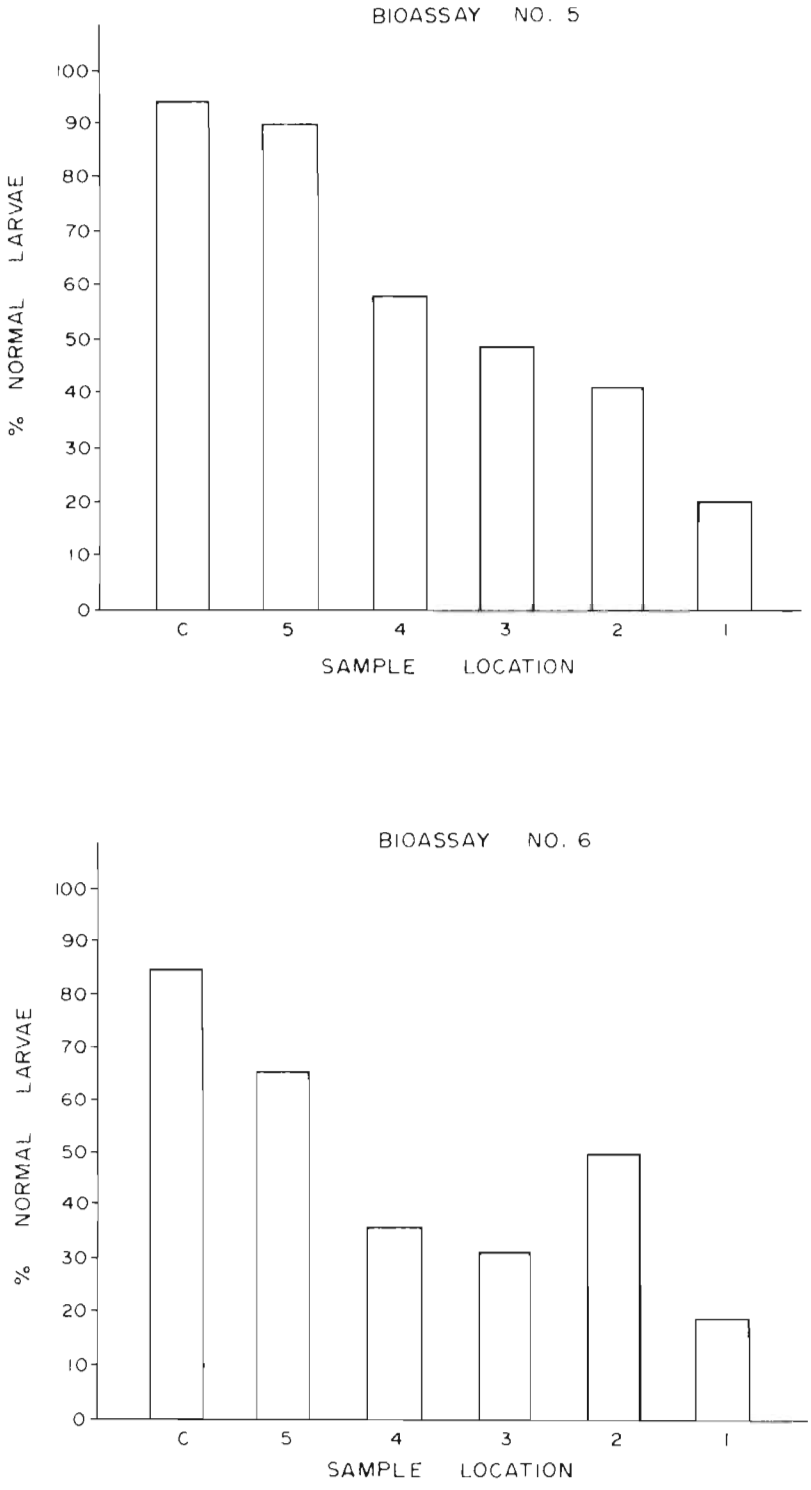


Fig. 4. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, March 19 above and April 11, 1979 below.

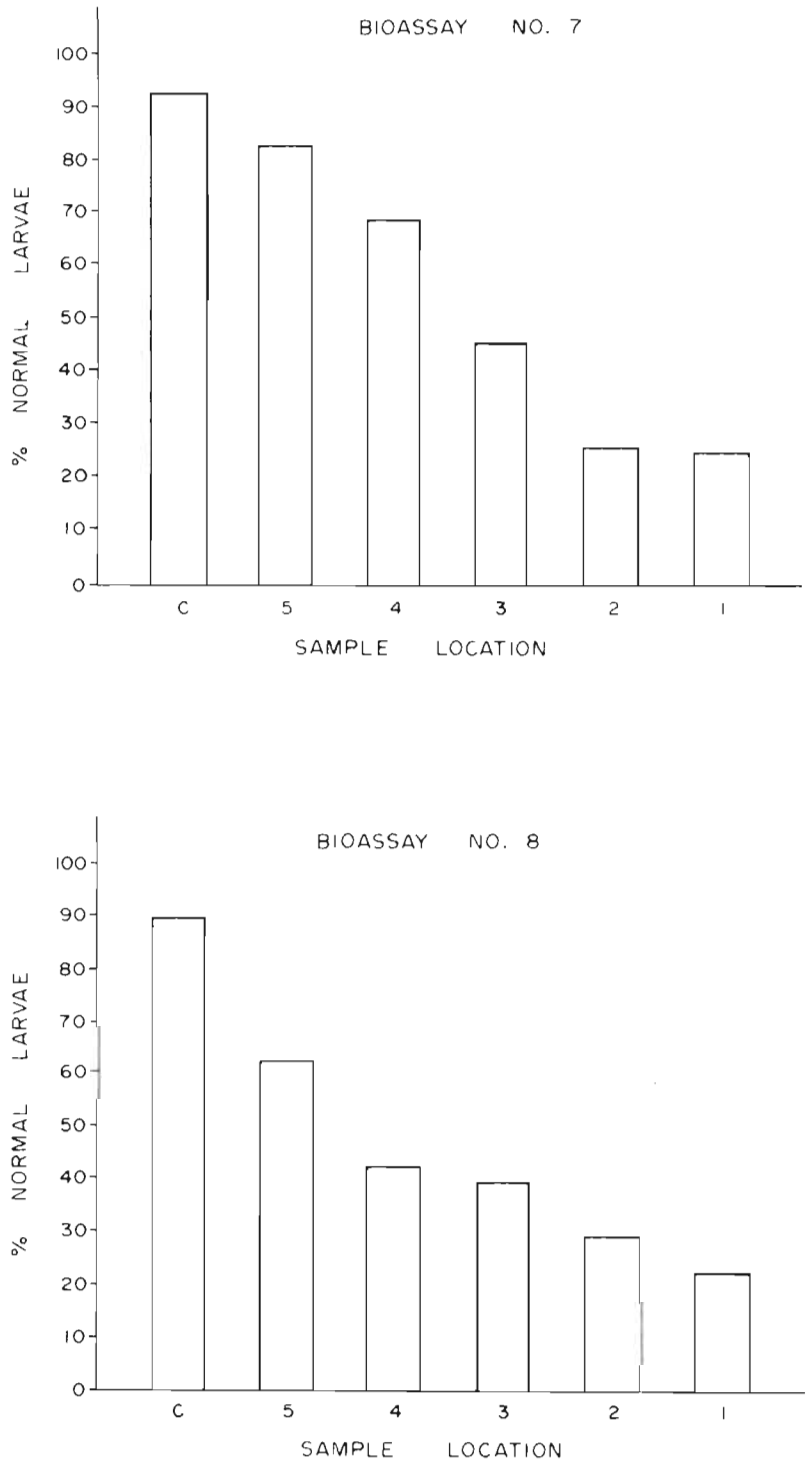


Fig. 5. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, April 23 above and April 30, 1979 below.

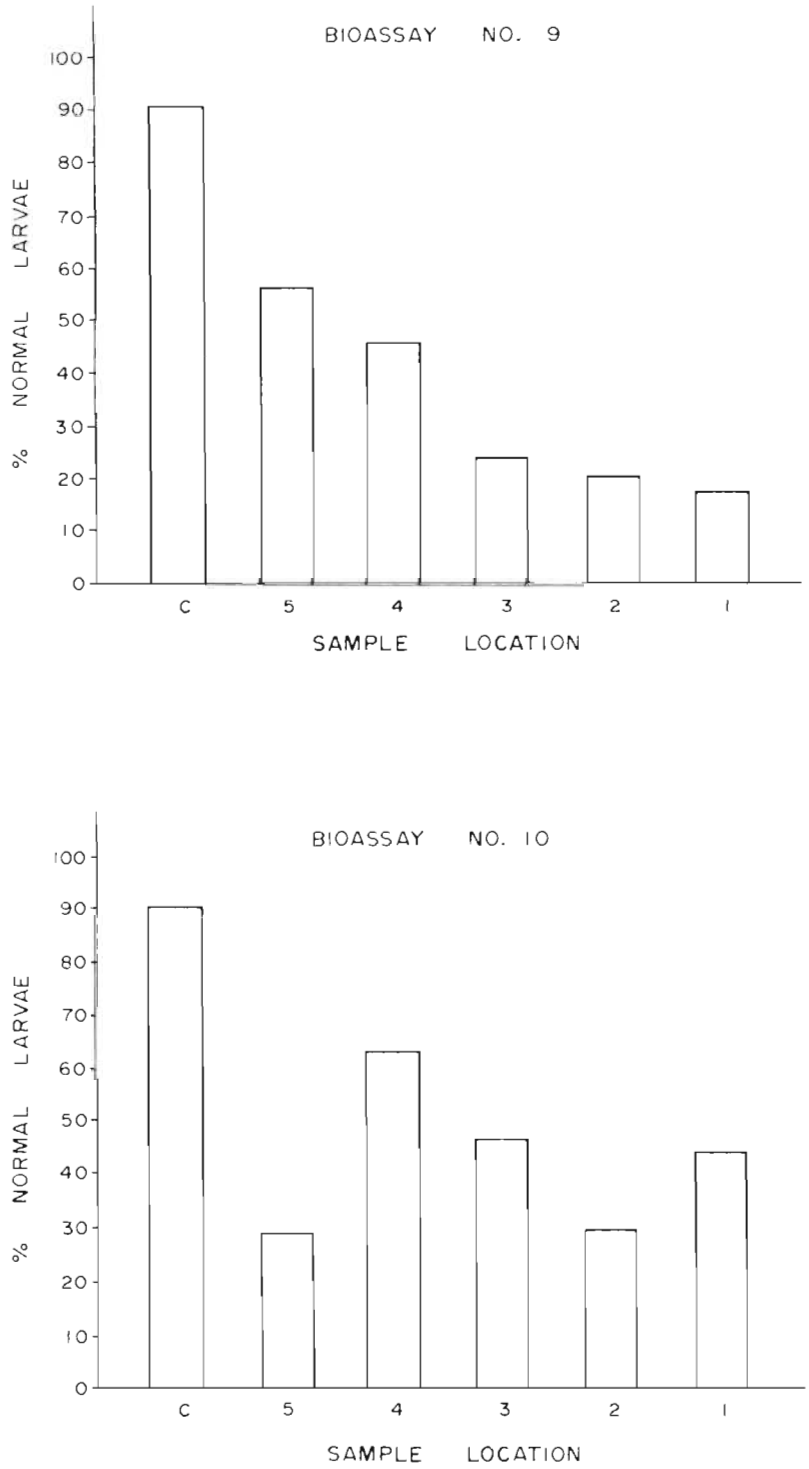


Fig. 6. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, May 14 above and May 29, 1979 below.

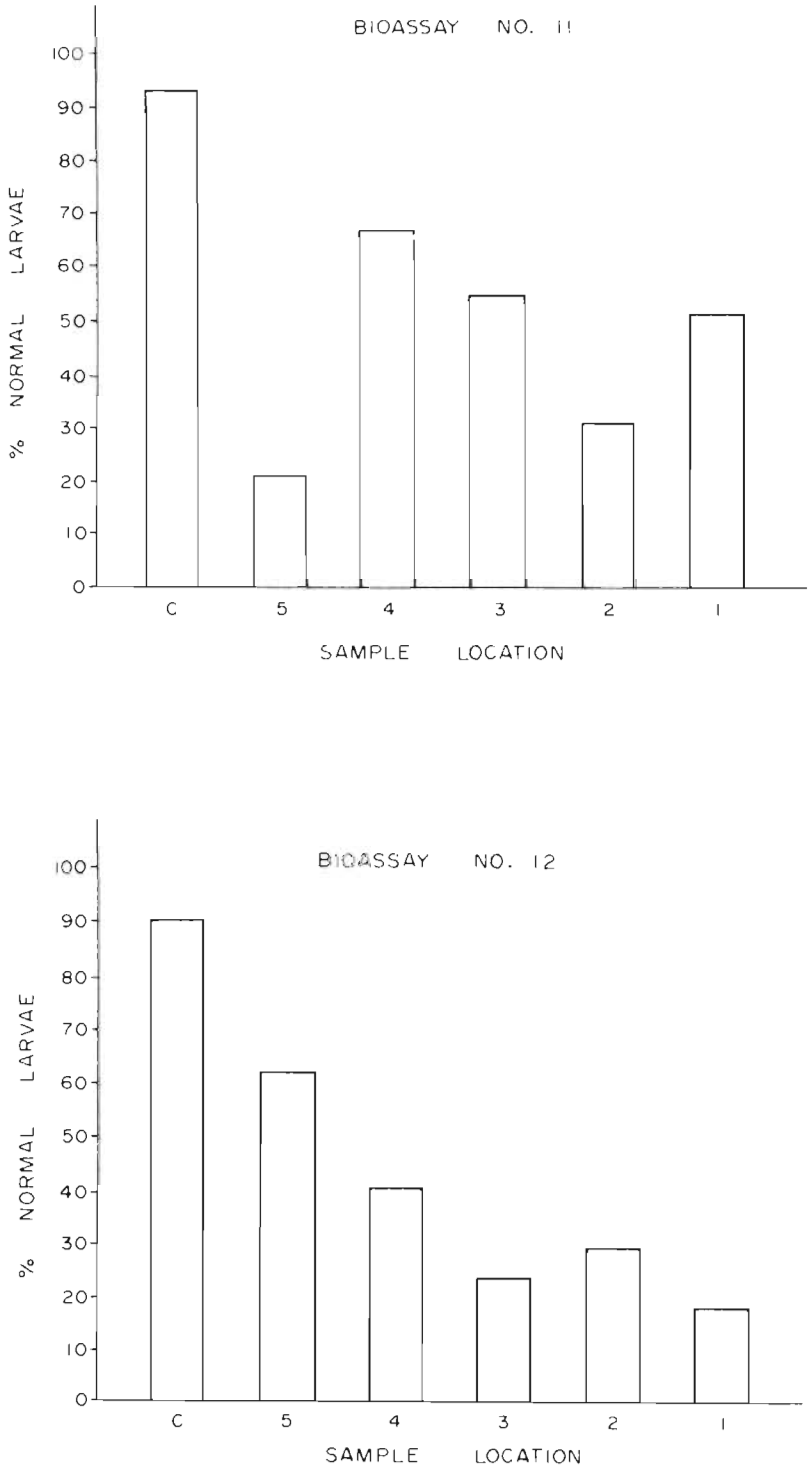


Fig. 7. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, June 4 above and June 20, 1979 below.

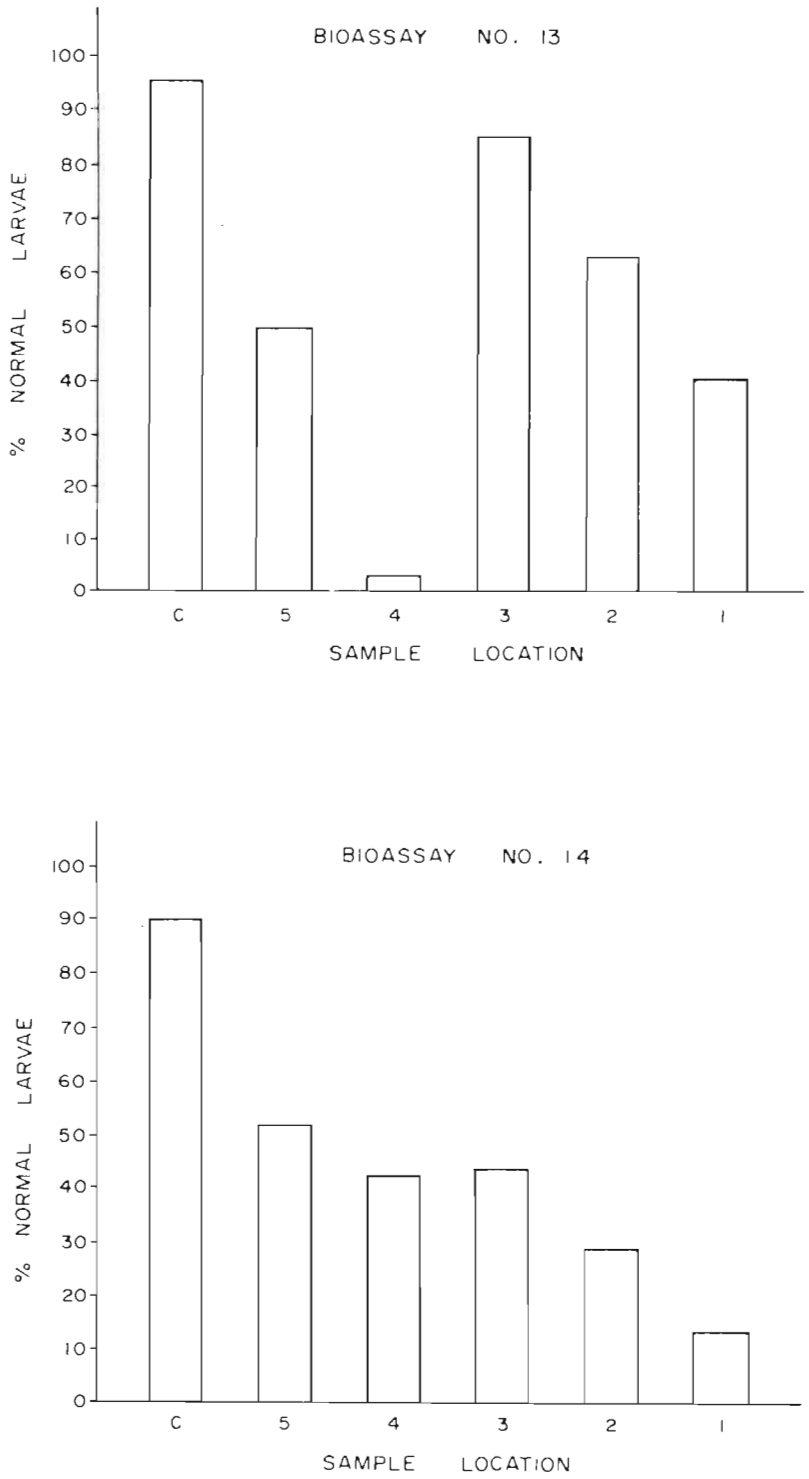


Fig. 8. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, July 6 above and July 18, 1979 below.

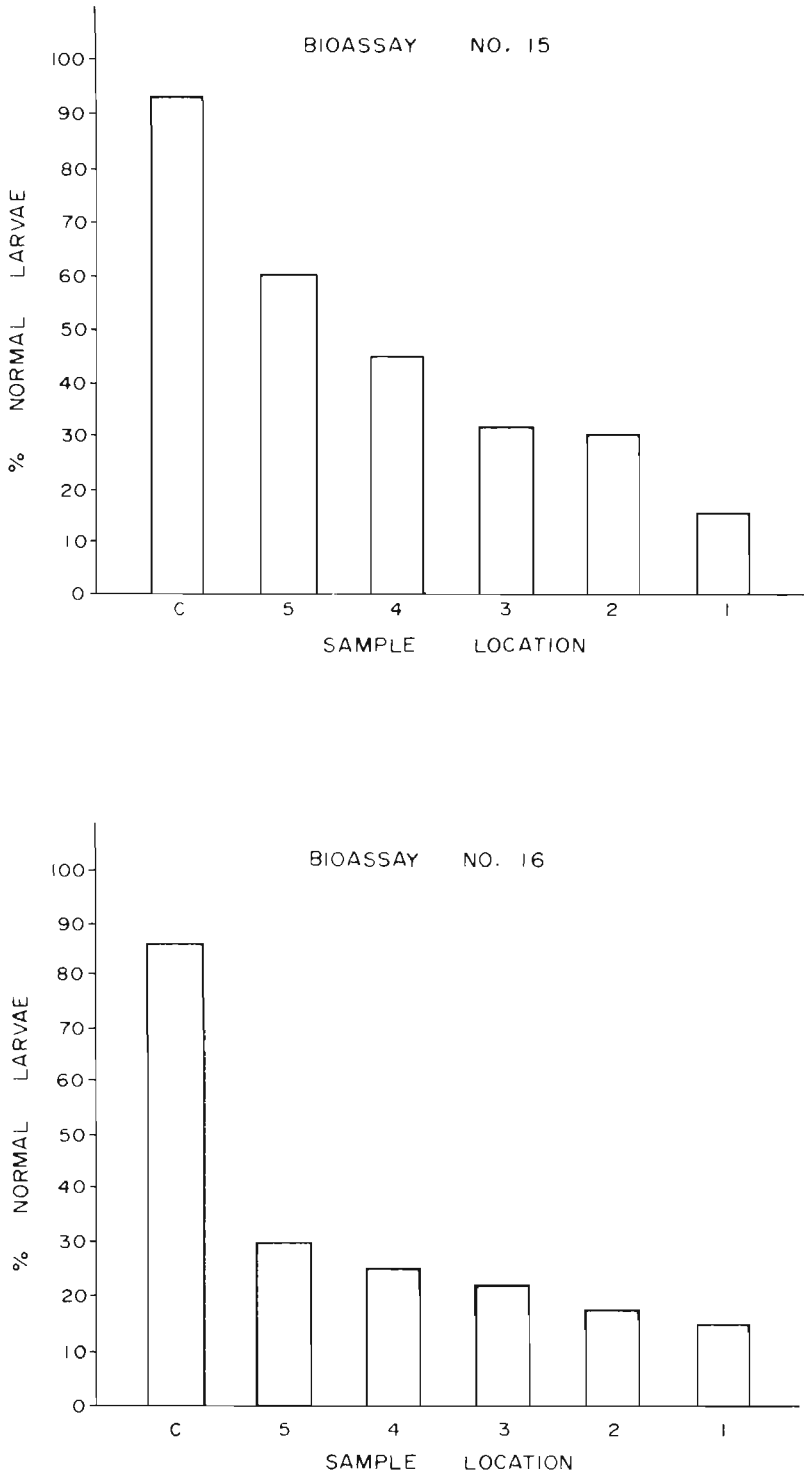


Fig. 9. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, July 31 above and August 7, 1979 below.

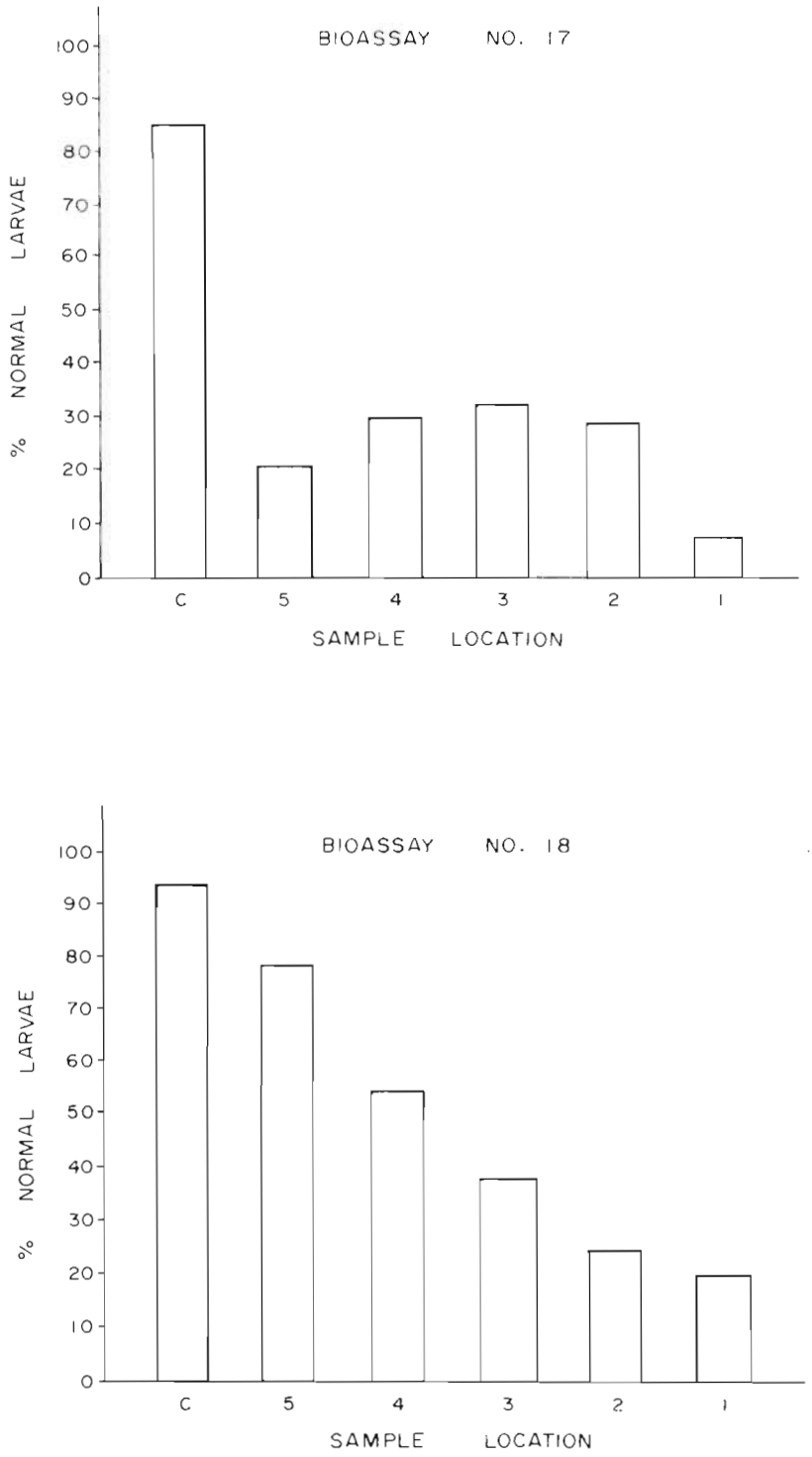


Fig. 10. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, August 14 above and August 20, 1979 below.

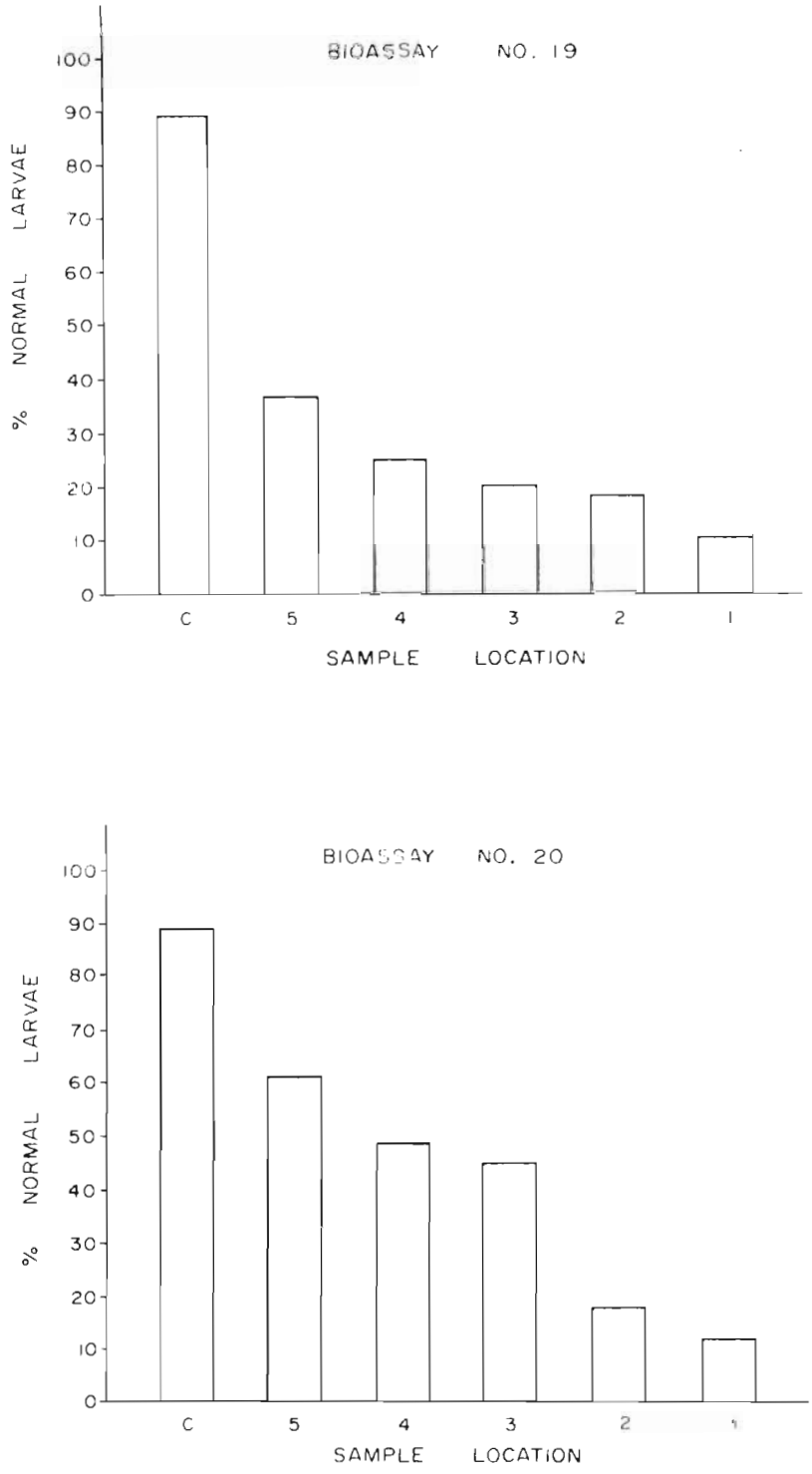


Fig. 11. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, August 27 above and September 10, 1979 below.

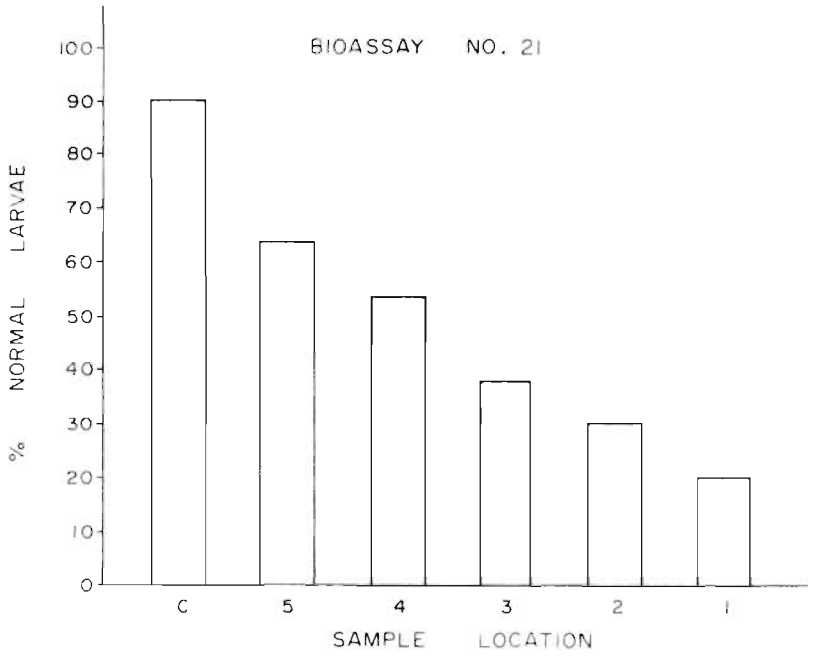


Fig. 12. Percent normal C. gigas larvae in control and water from five locations in Ladysmith Harbour, September 17, 1979.

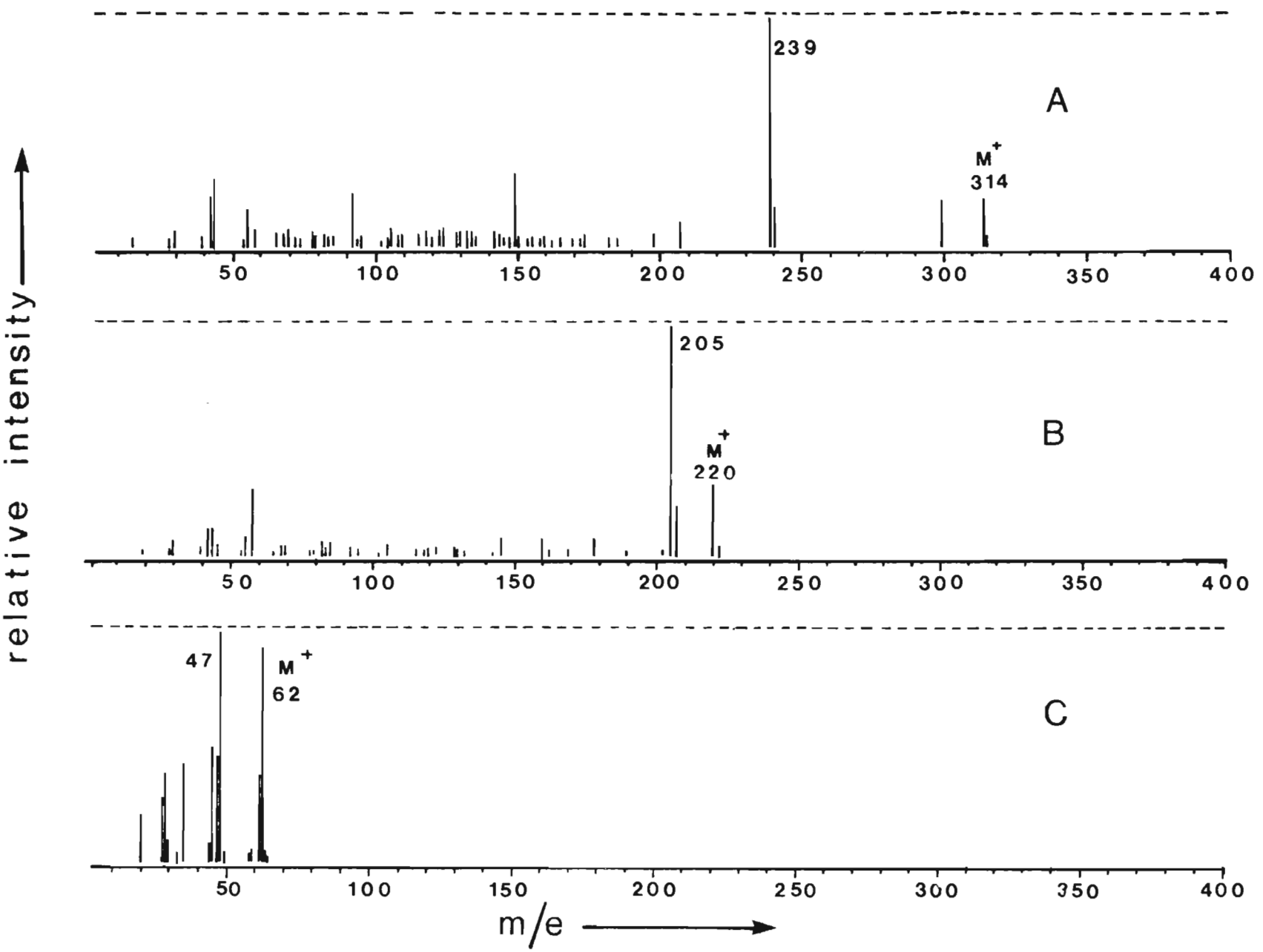


Fig. 13. A Mass spectrum of methyl dehydroabietate
B Mass spectrum of 2,6-diteriarybutyl-4-methylphenol
C Mass spectrum of dimethyl sulphide

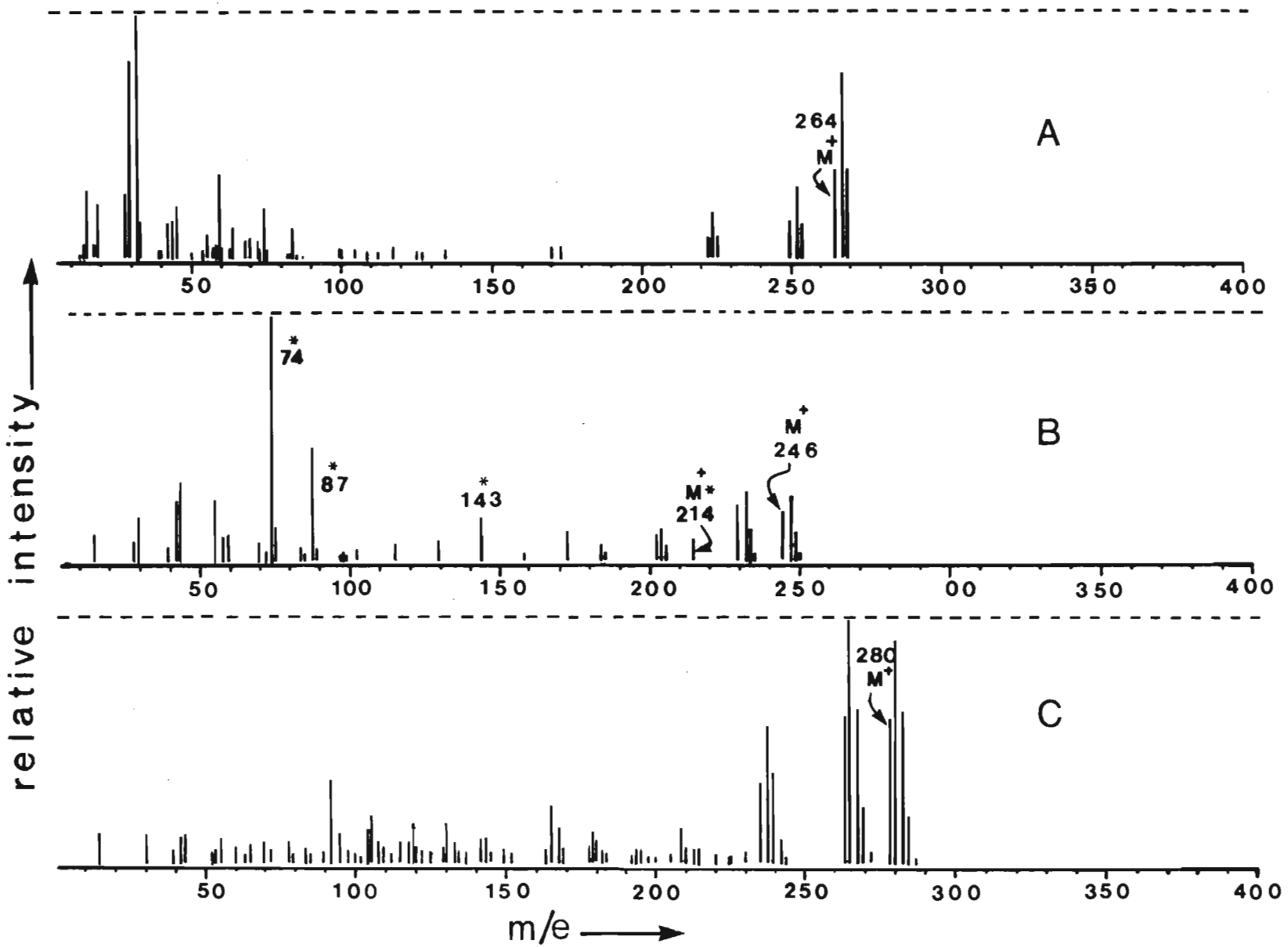


Fig. 14. A Mass spectrum of dibromoanisole
B Mass spectrum of a mixture of tetrachloroanisole and methyl laurate
C Mass spectrum of pentachloroanisole

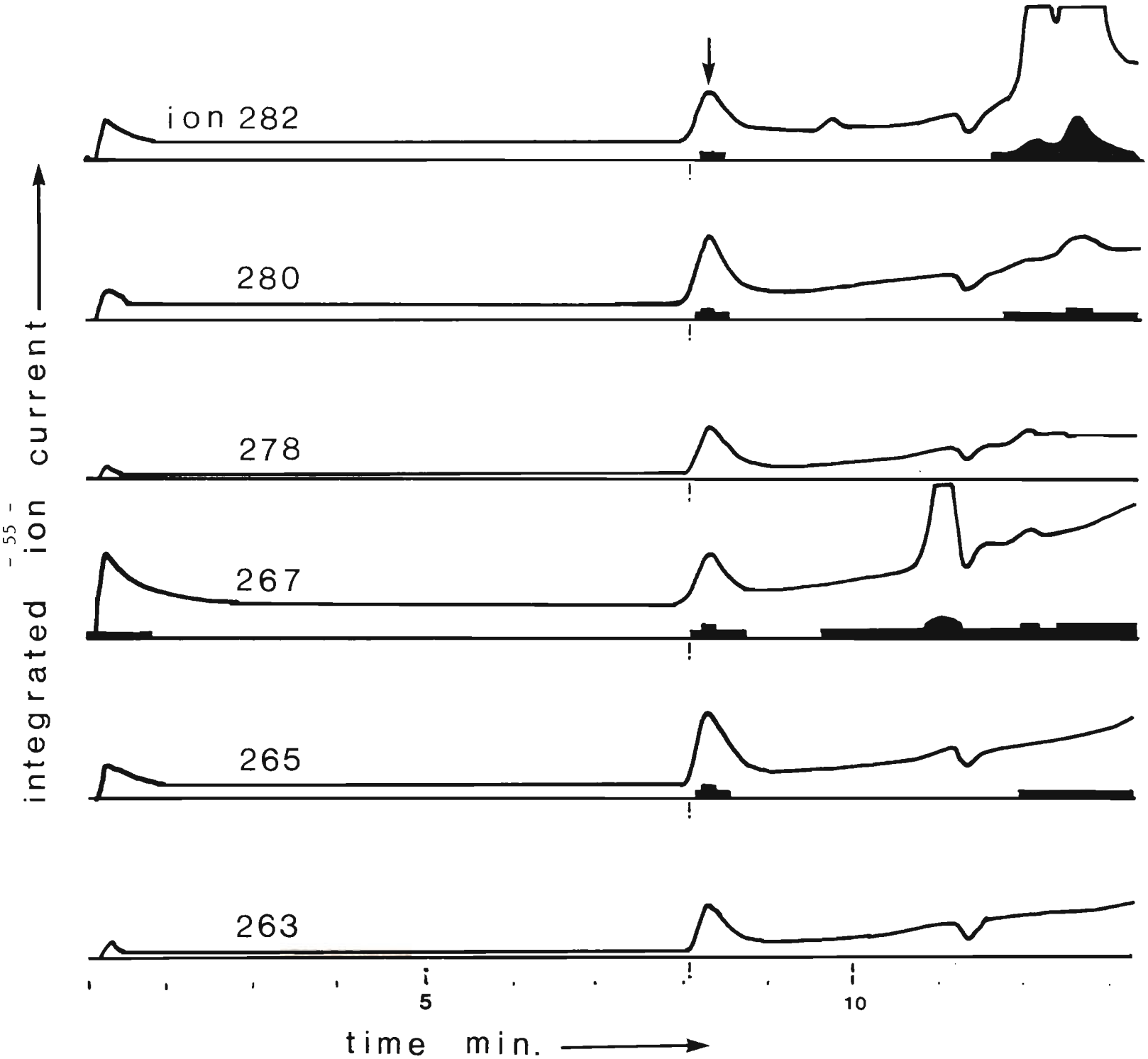


Fig. 15. SIM scan of an extract prepared from a foam sample in a search for pentachloroanisole.

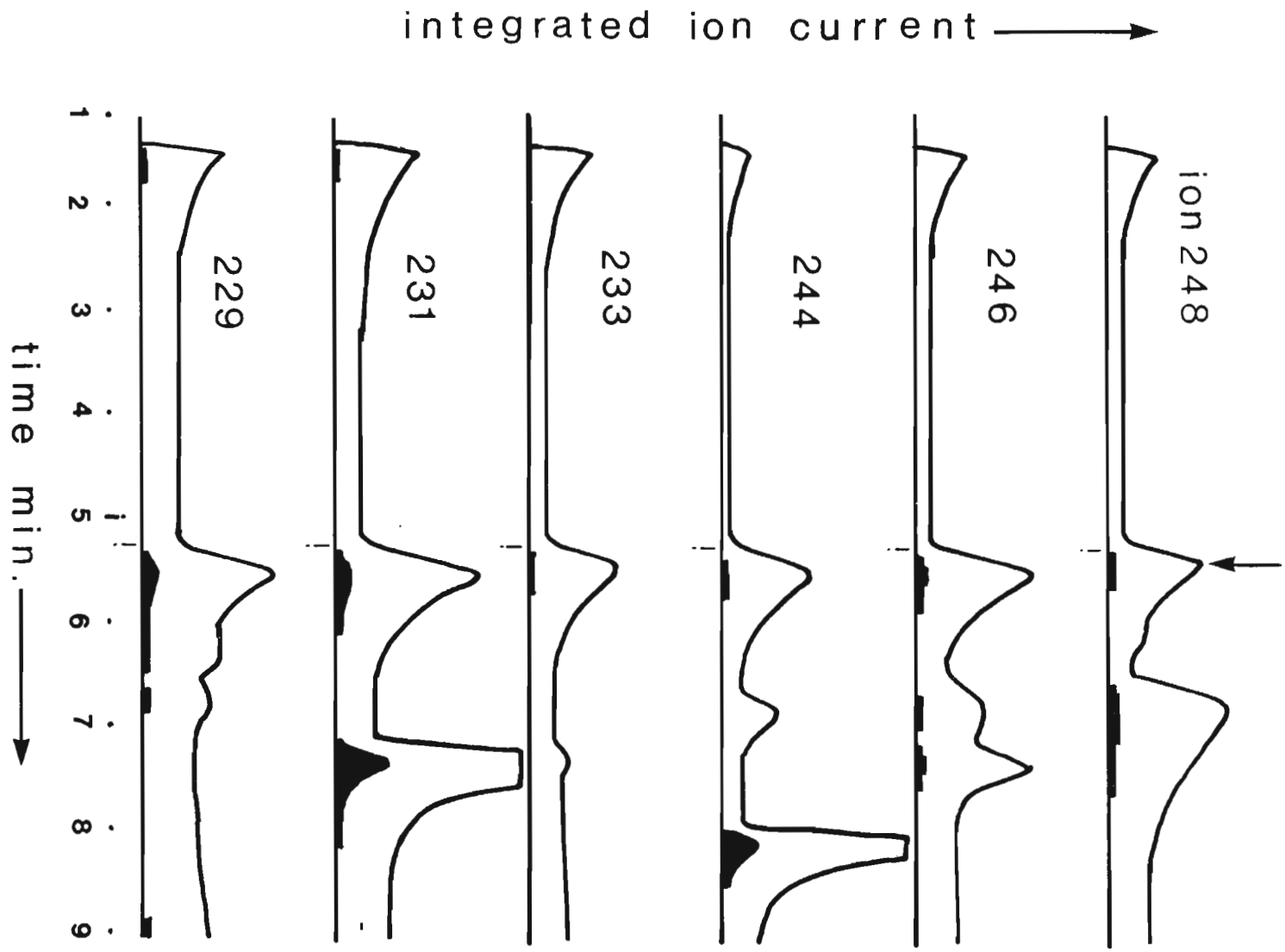


Fig. 16. SIM scan of an extract prepared from a foam sample in a search for tetrachloroanisole.